QUANTUM MECHANICS 3

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Lecture 1. State vectors in Hilbert space

Here we learn that the state of a quantum particle can be represented as a vector in Hilbert space. We use Dirac notations for these vectors and introduce the inner (scalar) product in Hilbert space ("bra(c)ket") in terms of "ket"- and "bra"-vectors.

1.1 State of a quantum particle

A classical state of a particle with f degrees of freedom ($f = 3$ for a particle without internal structure in a usual 3D space) is described by f coordinates $\{x_i\} \equiv (x_1, x_2, \ldots, x_f)$ and f (generalised) velocities $(\dot{x}_1, , \dot{x}_2, \ldots, \dot{x}_f)$ or, equivalently to the velocities, momenta $(p_1, p_2, \ldots, p_f)$. Therefore a state of such a classical particle is given by $2f$ numbers in total.

We know that in the quantum world simultaneous *exact* knowledge of particle's coordinates and momenta is impossible due to Heisenberg Uncertainty Principle. Moreover, although we can measure particle's coordinates with an arbitrary accuracy (losing information about momenta in the process), we can predict the results of measurements only *probabilistically* even when we know the state of the particle (e.g., its wavefunction) prior to the measurement. All available probabilistic information about coordinates and momenta of a quantum particle is contained in its wave function, $\Psi(\{x_i\})$, that fully characterises its *quantum state* at a given time, say $t{=}0$. The square of the wave function, $|\Psi(x_1, x_2, \ldots, x_f)|^2$, determines the statistical distribution of random coordinates: e.g., the probability that a particle moving in one dimension ($f = 1$) is found within the interval $[a, b]$ is given by

$$\mathrm{Prob}[a < x < b] = \int_a^b |\Psi(x)|^2 \, \mathrm{d}x \, . \tag{1.1}$$

Similarly, the momenta are also probabilistic quantities in QM. Although we don't know how to infer their distribution from the wave function,[1] some properties, like the average momentum, $\langle p \rangle$, or its variance, $\langle p^2 \rangle - \langle p \rangle^2$, are calculated for a given state using the operator $\hat{p} = -\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}x}$, so that

$$\langle p \rangle = \int_a^b \Psi^*(x) \hat{p} \Psi(x) \, \mathrm{d}x = \int_a^b \Psi^*(x) \left(-\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}x} \right) \Psi(x) \, \mathrm{d}x \tag{1.2}$$

$$\langle p^2 \rangle = \int_a^b \Psi^*(x) \hat{p}^2 \Psi(x) \, \mathrm{d}x = \int_a^b \Psi^*(x) \left(-\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}x} \right)^2 \Psi(x) \, \mathrm{d}x \, . \tag{1.3}$$

The time evolved wavefunction, $\Psi(x, t)$, obeys (as you know) the *linear* time-dependent Schrödinger equation, where the Hamiltonian $\hat{H}$ is the operator of full energy of the system:

$$\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}t} \Psi(x, t) = \hat{H} \Psi(x, t) \, . \tag{1.4}$$

The principal property of this equation is linearity: if $\Psi_1(x, t)$ and $\Psi_2(x, t)$ are its solutions, so it is their linear combination, $\alpha\Psi_1(x, t) + \beta\Psi_2(x, t)$ with any complex coefficients α and β. You know that many fundamental differential equations in physics are also linear, e.g., Maxwell's equations of electrodynamics, and solutions of the latter are electric and magnetic *vector* fields $\boldsymbol{E}$ and $\boldsymbol{B}$. In

[1] we will learn how to do it very soon

general, the linearity is the defining property of a vector space. Hence, it is tempting to say that solutions to the Schrödinger equations are vectors in a certain vector space.

This new definition of quantum states as *vectors* was introduced by Werner Heisenberg (1901-1976) and Paul Dirac (1902-1984). Familiar vectors in three-dimensional space are given by a set of three components, real (velocity, acceleration) or complex (electric and magnetic fields). Although vectors are geometric invariants, their components depend on the choice of coordinates (reference frame). Wavefunctions $\Psi(x)$ in quantum mechanics can be associated with state vectors which, in general, have *infinitely many complex components*: we shall see that values of $\Psi(x)$ at different points could be considered as components of the state vector in an infinitely dimensional vector space. Such a vector space that exists for any particular Hamiltonian $\hat{H}$ is called **Hilbert space**,[2] provided that the inner product of state vectors is defined.

1.2 Dirac notations

We know that vectors appear under different guises in maths and physics. The most simple but fundamental way to denote a vector is to draw an arrow. This gives a nice representation of physical quantities like displacements or velocities which have both magnitude (represented by the length of the arrow) and direction. It is also useful to give a name to such a vector, like $\vec{v}$ or $\boldsymbol{v}$.

To manipulate vectors, for example add and subtract them one can proceed in two ways. One is to use geometry by combining arrows as shown in Figure 1, or to use their components to perform calculations. In the latter case the sum of two vectors $\vec{a} = (a_1, a_2, a_3)$, $\vec{b} = (b_1, b_2, b_3)$ is $\vec{a} + \vec{b} = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$.

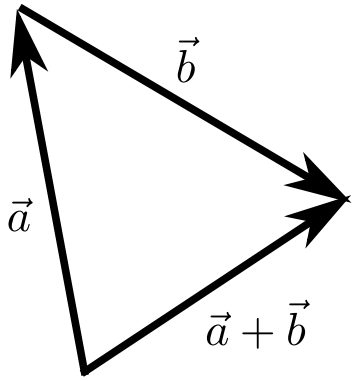


Figure 1: Vectors $\vec{a}$, $\vec{b}$ and their sum.

Both these two methods have their advantages and disadvantages. Manipulating arrows is very visual and intuitive, however it requires very good graphical skills and can be used only with given vectors. Using components allows perform general calculations when vectors are functions of time or other parameters. However, one should remember that, while a vector is geometric invariant, its components depend on the choice of coordinates (basis): the *same* vector will have *different* components if a different set of coordinates x, y, z (basis vectors) is chosen. The ability to use different bases is hugely advantageous for solving physical problems, including quantum mechanical ones as we shall see. (Note in passing that using different bases – frames – is one of the milestones of the Theory of Relativity, including Electrodynamics).

While many, if not all, physical calculations require choosing a particular basis at some stage, the geometric method, which is completely independent of the choice of basis, is extremely useful

[2]The mathematics of Hilbert space is a standard material and can be found in any decent book on linear algebra. A very good and concise exposition of it can be found in Chaper 1 of Shankar's book and I encourage everyone to use it as a reference for mathematical apparatus of Quantum Mechanics. It also provides a motivation to **Dirac notations** for vectors in Hilbert space. However, the minimal necessary information is given here.

for manipulations with vectors that can be added, subtracted, or scalar-multiplied; if needed, their components can be obtained at any stage by choosing an appropriate basis and projecting the vectors along the corresponding axes.

Such a basis-independent geometric approach to usual three-dimensional vectors has motivated Dirac notations for vectors in the Hilbert space. Of course it is impossible to draw arrows in infinite dimensional space (even for three dimensions it is difficult as paper has only two). However one can still introduce convenient notations, superseding those like $\vec{a}$ or $\boldsymbol{a}$ for usual three- (or two-) dimensional vectors in real spaces. The Dirac notations for a vector in Hilbert space became paramount:

$$\big|\Psi\big\rangle \tag{1.5}$$

pronounced "ket psi", uses (not necessarily, but often) Greek letters placed between vertical bar and the right angle brac**ket**. Despite using the same Greek letter Ψ as for the wave function in the beginning of this lecture the state vector $\big|\Psi\big\rangle$ and wave function $\Psi(x)$ are related but different objects: there is no reference to coordinate x in the former, while the latter is a representation of the state in the coordinate basis.

To summarise, we formulate the first QM postulate:

Postulate I. The state of the quantum particle is represented by vector $\big|\Psi\big\rangle$ in a Hilbert Space.

1.3 Properties of vectors in Hilbert space

Ket-vectors (or simply kets) in Hilbert space can be added together and multiplied by *complex* numbers. The result is also a vector. This constitutes the property of

Linearity: if $\big|\Psi\big\rangle$ and $\big|\Phi\big\rangle$ are vectors in a Hilbert space, then for any complex numbers $\alpha, \beta \in \mathbb{C}$ the linear combination $\alpha\big|\Psi\big\rangle + \beta\big|\Phi\big\rangle$ is also a vector in the same Hilbert space.

In particular $\big|\Psi\big\rangle - \big|\Psi\big\rangle = 0\big|\psi\big\rangle = \big|0\big\rangle$ is a unique zero ket-vector and often we will write simply $\big|\Psi\big\rangle - \big|\Psi\big\rangle = 0$.

Linear independence: Vectors $\big|\Psi_1\big\rangle$, $\big|\Psi_1\big\rangle$, ..., $\big|\Psi_j\big\rangle$, ... are linearly independent if the condition

$$\alpha_1\big|\Psi_1\big\rangle + \alpha_2\big|\Psi_2\big\rangle + \ldots + \alpha_k\big|\Psi_k\big\rangle + \ldots = \sum_{j=1}^{\infty} \alpha_j\big|\Psi_j\big\rangle = 0 \tag{1.6}$$

implies *all* the coefficients in this linear combination are zero, $\alpha_j = 0$.

A set $\big|\eta_n\big\rangle$, $n = 1, 2, \ldots$ of linearly independent vectors can be used as a *basis* so that any vector in the Hilbert space can be written as a linear combination,

$$\big|\Psi\big\rangle = \alpha_1\big|\eta_1\big\rangle + \alpha_2\big|\eta_2\big\rangle + \alpha_2\big|\eta_2\big\rangle + \ldots = \sum_{j=1}^{\infty} \alpha_j\big|\eta_j\big\rangle\,. \tag{1.7}$$

One can say that state $\big|\Psi\big\rangle$ is a quantum superposition of states $\big\{\big|\eta_j\big\rangle\big\}$.

In order to be able to represent *any* vector in this way, the basis must be *complete*.[3] The criterion of completeness of a basis will be given later.

The complex numbers α_1, α_2,... are called *components* of vector $\big|\Psi\big\rangle$ in the basis $\big\{\big|\eta_j\big\rangle\big\}$.

Show that the expansion (1.7) is unique in a given basis.

1.4 Scalar product in Hilbert Space. Bra and ket vectors.

Any ket-vector can be represented as a linear combination (1.7). The complex numbers α_n are components of $\big|\Psi\big\rangle$ in the basis $\big|\eta_n\big\rangle$. In standard three-dimensional space spanned by ***orthonormal*** (*i.e.*, all of unit length and orthogonal to each other) basis vectors $\hat{e}_1, \hat{e}_2, \hat{e}_3$ we find the components of a vector $\vec{a} = a_1\hat{e}_1 + a_2\hat{e}_2 + a_3\hat{e}_3$ by taking scalar products: $a_j = \hat{e}_j \cdot \vec{a}$. A similar procedure is implemented in Hilbert space, where we need to define the inner (a.k.a. scalar) product, of two vectors denoted by

$$\langle\Psi \,|\, \Phi\rangle$$

(pronounced "bra-ket psi phi").

Unlike the case of real vectors, *the order of arguments in the inner product does matter*, and the scalar $\langle\Psi \,|\, \Phi\rangle$ is, in general, a complex number. The first property (axiom) of the inner product is

Skew-symmetry: $\langle\Psi \,|\, \Phi\rangle = \langle\Phi \,|\, \Psi\rangle^*$.

This ensures that "square length" of a vector, $\langle\Psi \,|\, \Psi\rangle$, is a real number. For the QM applications we demand this number to be non-negative,[4] so that the following property holds

Positive semidefiniteness: $\langle\Psi \,|\, \Psi\rangle \geq 0$,

where zero is obtained only for zero vector. The third property is

Linearity in ket: $\big\langle\Psi\big|\Big(\alpha\big|\Phi_1\big\rangle + \beta\big|\Phi_2\big\rangle\Big) \equiv \langle\Psi \,|\, \alpha\Phi_1 + \beta\Phi_2\rangle = \alpha\langle\Psi \,|\, \Phi_1\rangle + \beta\langle\Psi \,|\, \Phi_2\rangle$.

for any two complex numbers α, β.

The inner product allows us to define *orthogonality* of vectors: $\big|\Psi\big\rangle$ and $\big|\Phi\big\rangle$ are orthogonal if their inner product vanishes. We can also define the *norm* or length of the vector $\big|\Psi\big\rangle$ by taking square root of its inner product with itself, $\|\Psi\| \equiv \sqrt{\langle\Psi \,|\, \Psi\rangle}$.

The inner product makes the new object, $\big\langle\Psi\big|$ called "bra-vector" or simply "bra", to appear. One can think about it as of a different type of vectors, dual to ket-vectors. In the more formal mathematical language bras are linear functionals, *i.e.* objects which act on ket vectors to produce a number with the help of the inner product we defined. The convenience is in the fact that we can deal with $\langle\Phi \,|\, \psi\rangle$ as, indeed, with a product where both the left and the right part are separate objects.

[3] As an example of an incomplete basis consider the following situation in the regular three-dimensional space: take two vectors lying in the xy plane. They do not form a complete basis in three dimensions because no linear combination of these two vectors can represent a vector not parallel to this plane (*i.e.* having a non-zero z component).

[4] This is not a general property of the inner product in a vector space. E.g., in Special Relativity the "square length" of a vector is always real but could be negative or zero.

It is important to distinguish bras from kets. They are related by the fact that both $\big|\Psi\big\rangle$ and $\big\langle\Psi\big|$ ***describe the same quantum state***; however, they live in different Hilbert spaces and cannot appear together in linear combinations, *i.e.* expressions like $\big|\Psi\big\rangle + \big\langle\Phi\big|$ make no sense!

Loosely (at the moment – a strict definition will follow), the ket vector corresponds to the QM wave function, and the bra vector – to its conjugate.

It follows from skew-symmetry that

$$\begin{aligned}\langle \alpha\Phi_1 + \beta\Phi_2 \,|\, \Psi\rangle &= \langle \Psi \,|\, \alpha\Phi_1 + \beta\Phi_2\rangle^* \\ &= \alpha^*\langle \Psi \,|\, \Phi_1\rangle^* + \beta^*\langle \Psi \,|\, \Phi_2\rangle^* \\ &= \alpha^*\langle \Phi_1 \,|\, \Psi\rangle + \beta^*\langle \Phi_2 \,|\, \Psi\rangle\,.\end{aligned} \tag{1.8}$$

This property is called *antilinearity in bra*. It can be seen as a rule to produce a bra-vector:

$$\big\langle \alpha\Phi_1 + \beta\Phi_2\big| = \alpha^*\big\langle \Phi_1\big| + \beta^*\big\langle \Phi_2\big| \tag{1.9}$$

from bras $\big\langle\Phi_1\big|$ and $\big\langle\Phi_2\big|$. One may ask: "What are they?", namely given a ket-vector $\big|\Psi\big\rangle$ how to make the corresponding[5] bra vector $\big\langle\Psi\big|$?

The simplest way to answer this question is to use some basis, say $\big|n\big\rangle$, $(n = 1, 2, \ldots)$ in the Hilbert space of ket-vectors. We will often omit letters for basis vectors. From now on we always assume that the basis is ***orthonormal***

$$\langle n \,|\, m\rangle = \delta_{nm} = \begin{cases} 1, & n = m \\ 0, & n \neq m \end{cases}\,. \tag{1.10}$$

Using this formula we can associate the bra-vector $\big\langle n\big|$ to each basis ket-vector $\big|n\big\rangle$. Now write a linear expansion of a ket-vector:

$$\big|\Psi\big\rangle = \alpha_1\big|1\big\rangle + \alpha_2\big|2\big\rangle + \ldots = \sum_{m=1}^{\infty} \alpha_m\big|m\big\rangle \tag{1.11}$$

and the corresponding bra-vectors follows from the antilinearity:

$$\big\langle\Psi\big| = \alpha_1^*\big\langle 1\big| + \alpha_2^*\big\langle 2\big| + \ldots = \sum_{m=1}^{\infty} \alpha_m^*\big\langle m\big|\,. \tag{1.12}$$

To extract the component α_n of $\big|\Psi\big\rangle$ in a given basis we take the inner product:

$$\langle n \,|\, \Psi\rangle = \sum_{m=1}^{\infty} \alpha_m \langle n \,|\, m\rangle = \sum_{m=1}^{\infty} \alpha_m \delta_{nm} = \alpha_n\,. \tag{1.13}$$

Given another vector expanded in the same basis,

$$\big|\Phi\big\rangle = \beta_1\big|1\big\rangle + \beta_2\big|2\big\rangle + \ldots = \sum_{m=1}^{\infty} \beta_m\big|m\big\rangle \tag{1.14}$$

[5] mathematicians use word "adjoint" or "conjugate"

we can calculate the inner product,

$$\langle \Psi \,|\, \Phi \rangle = \sum_{m=1}^{\infty} \alpha_m^* \langle m| \sum_{m'=1}^{\infty} \beta_{m'} |m'\rangle = \sum_{m=1,m'=1}^{\infty} \alpha_m^* \beta_{m'} \langle m \,|\, m' \rangle \tag{1.15}$$

$$= \sum_{m=1,m'=1}^{\infty} \alpha_m^* \beta_{m'} \delta_{mm'} = \sum_{m=1}^{\infty} \alpha_m^* \beta_m . \tag{1.16}$$

In particular, the vector norm is

$$\langle \Psi \,|\, \Psi \rangle = \sum_{m=1}^{\infty} \alpha_m^* \alpha_m = \sum_{m=1}^{\infty} |\alpha_m|^2 . \tag{1.17}$$

Let us stress that not only $|\Psi\rangle$ and $\langle\Psi|$ but also $|\alpha\Psi\rangle$ and $\langle\alpha\Psi|$ represent the same quantum state, albeit different vectors in the appropriate Hilbert space.

To narrow down the representation of quantum state, we introduce the normalised states,

$$|\widetilde{\Psi}\rangle \equiv \frac{|\Psi\rangle}{\sqrt{\langle\Psi|\,\Psi\rangle}}. \tag{1.18}$$

This still does not make the representation unique as $|\mathrm{e}^{i\phi}\widetilde{\Psi}\rangle$ is also normalised for any real number ϕ. Moreover, dealing with normalised states at any intermediate stage of calculations is not convenient since, *e.g.*, the sum of normalised states is not necessarily normalised.

A ket-vector can be defined in terms of its components in a given basis that can be arranged in a form of a matrix column:

$$|\Psi\rangle \leftrightarrow \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \end{pmatrix} \qquad |\Phi\rangle \leftrightarrow \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \end{pmatrix} . \tag{1.19a}$$

Here $\leftrightarrow$ means correspondence within a given basis.[6] The corresponding bra-vectors are then obtained by Hermitian conjugation, i.e. by transposing and taking complex conjugate:

$$\langle\Psi| \leftrightarrow \left(\alpha_1^*, \alpha_2^*, \ldots\right) \qquad \langle\Phi| \leftrightarrow \left(\beta_1^*, \beta_2^*, \ldots\right) . \tag{1.19b}$$

The inner product becomes the standard matrix multiplication:

$$\langle \Psi \,|\, \Phi \rangle = \left(\alpha_1^*, \alpha_2^*, \ldots\right) \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \end{pmatrix} = \alpha_1^* \beta_1 + \alpha_2^* \beta_2 + \ldots . \tag{1.20}$$

It is very useful to remember this correspondence between state vectors (bras and kets) and the corresponding row and column vectors, but we should always bear in mind that this correspondence relies on the choice of basis (as usual when one uses components to define a vector). So the matrix expressions in (1.19) give *representations* of the ket or bra state vectors in a given basis. If we have chosen a different basis, $|\tilde{n}\rangle$ we would end up with different components $\alpha_{\tilde{n}}$,$\beta_{\tilde{n}}$ for the vectors $|\Psi\rangle$, $|\Phi\rangle$. Of course the inner product is invariant under such a change, and the inner product in Dirac notations, $\langle \Psi \,|\, \Phi \rangle$, is truly basis-independent.

[6] We shall formalise this correspondence in Lecture 2.

Lecture 2. Operators in Hilbert space. 2nd and 3rd QM postulates.

Here we introduce operators in Hilbert space acting on ket- and bra-vectors and give both their matrix representation in a particular basis and the Dirac basis-independent representation. This allows us to define self-adjoint (Hermitian) operators representing observables, and unitary operators corresponding to allowed transformations.. As this is a revision of material given in Y2 Maths, I give it here for completeness but will scant over this part in the class, focusing on the QM postulates of measurements.

2.1 Linear operators

Once we have established the description of quantum mechanical states as vectors in Hilbert space we move to the next stage and want to manipulate them in order to be able to describe quantum dynamics. For this we need *operators*, *i.e.* rules to transform vectors into one another. We know examples of operators in regular three-dimensional Euclidean space like rotations. In Hilbert space operators transform state vectors like this:

$$\big|\Psi\big\rangle \xrightarrow{\hat{O}} \big|\Phi\big\rangle\,. \tag{2.1}$$

We will write $\big|\Phi\big\rangle = \hat{O}\big|\Psi\big\rangle = \big|\hat{O}\Psi\big\rangle$ which means that operator $\hat{O}$ acts on the state-vector $\big|\Psi\big\rangle$ and produces Φ. Operators are denoted by small hats, but sometimes these are omitted if there is no confusion.

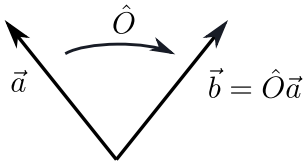


Figure 2: Rotation.

We will only deal with *linear* operators which take linear superpositions into linear superpositions of transformed states:

$$\hat{O}\big|\alpha\Psi_1 + \beta\Psi_2\big\rangle = \alpha\hat{O}\big|\Psi_1\big\rangle + \beta\hat{O}\big|\Psi_2\big\rangle = \alpha\big|\Phi_1\big\rangle + \beta\big|\Phi_2\big\rangle\,, \tag{2.2}$$

for two complex numbers α, β and $\hat{O}\big|\Psi_{1,2}\big\rangle = \big|\Phi_{1,2}\big\rangle$. Operators can also act on bra-vectors and we will return to this point later.

The simplest, but very important example of a linear operator is the identity operator $\hat{I}$, which leaves the state-vector unchanged: $\hat{I}\big|\Psi\big\rangle = \big|\Psi\big\rangle$. We have seen that any orthonormalised complete basis $\{\eta_n\}$ satisfies $\sum_n \big|\eta_n\big\rangle\big\langle\eta_n\big| = \hat{I}$.

Operators can be combined into products:

$$\hat{O}_2\hat{O}_1\big|\Psi\big\rangle = \hat{O}_3\big|\Psi\big\rangle = \hat{O}_2\big|\hat{O}_1\Psi\big\rangle\,, \tag{2.3}$$

so that $\hat{O}_3 = \hat{O}_2\hat{O}_1$, *i.e.* operator $\hat{O}_3$ is the combined result of acting first with $\hat{O}_1$ and then with $\hat{O}_2$. {The order of the operators' action is *from right to left* like in Semitic languages, *i.e.* in the product $\hat{A}\hat{B}$ the operator $\hat{B}$ acts first. The order is important: in general, $\hat{O}_1\hat{O}_2 \neq \hat{O}_2\hat{O}_1$. A good example is two consecutive rotations in three dimensions around different axes.

The difference $\hat{O}_1\hat{O}_2 - \hat{O}_2\hat{O}_1 \equiv [\hat{O}_1, \hat{O}_2]$ is called the **commutator** of $\hat{O}_1$ and $\hat{O}_2$.

One defines the **inverse** operator $\hat{O}^{-1}$ such that $\hat{O}^{-1}\hat{O} = \hat{O}\hat{O}^{-1} = \hat{I}$. This operator reverts the action of $\hat{O}$. The **inverse of a product**

$$\left(\hat{O}_2\hat{O}_1\right)^{-1} = \hat{O}_1^{-1}\hat{O}_2^{-1} \tag{2.4}$$

since $\hat{I} = \left(\hat{O}_2\hat{O}_1\right)^{-1}\left(\hat{O}_2\hat{O}_1\right) = \hat{O}_1^{-1}\hat{O}_2^{-1}\hat{O}_2\hat{O}_1 = \hat{O}_1^{-1}\hat{O}_1$. In simple words one has to undo the action of the last operator that acted on the state.

2.2 Matrix and Dirac representation of operators

To specify an operator we must define how it transforms *any* vector in a Hilbert space. One way of doing this is to choose a complete ***orthonormal*** basis $\{|n\rangle\}$ and define the action of an operator $\hat{B}$ on each basis vector:

$$\hat{B}|n\rangle = |\widetilde{n}\rangle = \sum_j \beta_{jn}|j\rangle\,. \tag{2.5}$$

As any vector can be spanned in basis $\{|n\rangle\}$, we have expanded $|\widetilde{n}\rangle$ above. The expansion coefficients β_{jn} depend on two indices, the first is the summation index and the second indicates on which basis vector we have acted by $\hat{B}$. Now we multiply both sides of (2.5) by the bra-vector $\langle m|$ and use the orthonormality of basis $\{|n\rangle\}$:

$$\langle m\,|\,\hat{B}\,|\,n\rangle = \langle m|\sum_{j=1}^{\infty}\beta_{jn}|j\rangle = \sum_{j=1}^{\infty}\beta_{jn}\underbrace{\langle m\,|\,j\rangle}_{=\delta_{mj}} = \beta_{mn}, \tag{2.6}$$

as the Krönecker symbol δ_{mj} kills all the terms in the sum but that with $j = m$. We call complex numbers $\beta_{mn} = \langle m|\hat{B}|n\rangle$ **matrix elements** of the operator $\hat{B}$ in the basis $|n\rangle$. Obviously, the set $\{\beta_{mn}\}$ can be considered as a matrix, possibly infinitely-dimensional. Moreover, the operator actions can be represented by matrix multiplication.

To see this, let us take the representation of an arbitrary state $|\Psi\rangle$ in the same basis $\{|n\rangle\}$:

$$|\Psi\rangle = \sum_{n=1}^{\infty}\alpha_n|n\rangle \text{ with components } \alpha_n = \langle n\,|\,\Psi\rangle. \tag{2.7}$$

The transformed state $|\Phi\rangle = \hat{B}|\Psi\rangle$ is obtained by acting with the operator on each basis vector in the sum. Using (2.5), we find

$$|\Phi\rangle = \hat{B}\sum_{n=1}^{\infty}\alpha_n|n\rangle = \sum_{n=1}^{\infty}\alpha_n\hat{B}|n\rangle = \sum_{n=1}^{\infty}\alpha_n\sum_{j=1}^{\infty}\beta_{jn}|j\rangle = \sum_{j=1}^{\infty}\left(\sum_{n=1}^{\infty}\beta_{jn}\alpha_n\right)|j\rangle \equiv \sum_j \gamma_j|j\rangle, \tag{2.8}$$

where $\{\gamma_j = \langle j\,|\,\Phi\rangle\}$ is the column-matrix representation of $|\Phi\rangle$. Hence, indeed, we see the equivalence of the operator action to the matrix multiplication in the chosen basis:

$$|\Phi\rangle = \hat{B}|\Psi\rangle \quad\Leftrightarrow\quad \gamma_j = \sum_{n=1}^{\infty}\beta_{jn}\alpha_n \quad\Leftrightarrow\quad \begin{pmatrix}\gamma_1\\ \gamma_2\\ \vdots\end{pmatrix} = \begin{pmatrix}\beta_{11} & \beta_{12} & \dots\\ \beta_{21} & \beta_{22} & \dots\\ \vdots & \vdots & \ddots\end{pmatrix}\begin{pmatrix}\alpha_1\\ \alpha_2\\ \vdots\end{pmatrix}. \tag{2.9}$$

This we have established the correspondence between operators in Hilbert space and matrices. This correspondence relies on a particular choice of basis, albeit it holds in any complete orthonormal basis. Now we shall use Dirac notations for specifying operator actions in a basis-independent way.

Let us return to (2.7) and substitute the expression for α_n into the expansion:

$$\big|\Psi\big\rangle = \sum_{n=1}^{\infty} \alpha_n \big|n\big\rangle = \sum_{n=1}^{\infty} \langle n \,|\, \Psi\rangle \big|n\big\rangle = \sum_{n=1}^{\infty} \big|n\big\rangle\langle n \,|\, \Psi\rangle = \left(\sum_{n=1}^{\infty} \big|n\big\rangle\big\langle n\big|\right)\big|\Psi\big\rangle\,. \tag{2.10}$$

by moving the ket-vectors to the front. In this form it becomes clear that the right hand side is a result of an operator $\sum_i \big|n\big\rangle\big\langle n\big|$ acting on $\big|\Psi\big\rangle$. Since this operator acting on $\big|\Psi\big\rangle$ produces $\big|\Psi\big\rangle$ again, it is nothing but the identity operator:

$$\hat{I} = \sum_n \big|n\big\rangle\big\langle n\big|\,. \tag{2.11}$$

This complicated way to write the operator which does nothing is called the **resolution of unity**[1] and will become one of the most useful tricks in doing calculations with Dirac notations. I deliberately left out the limits in the summation sign: the resolution of unity works as long as basis $\big\{\big|n\big\rangle\big\}$, whether it comprises a finite or infinite number of basis vectors, is complete in a Hilbert space. Actually, this is nothing more than the criterion of completeness:

An orthonormal set, $\big\{\big|n\big\rangle\big\}$, is complete in a Hilbert space iff $\sum_n \big|n\big\rangle\big\langle n\big| = \hat{I}$

Let us use (2.11) to represent an operator $\hat{B}$ in a basis-independent Dirac form:

$$\hat{B} = \hat{I}\hat{B}\hat{I} = \left(\sum_i \big|i\big\rangle\big\langle i\big|\right)\hat{B}\left(\sum_j \big|j\big\rangle\big\langle j\big|\right) = \sum_{ij} \big|i\big\rangle\beta_{ij}\big\langle j\big|. \tag{2.12}$$

On the face of it, this expression still depends on the basis, but this is not the case: if the basis changed the matrix elements β_{ij} would change as well while the operator remains the same. We will come to this point later when we will discuss basis transformations.

Naturally, the Dirac representation of (2.12) is consistent with the matrix representation of (2.6). Indeed,

$$\big\langle m\,\big|\,\hat{B}\,\big|\,n\big\rangle = \big\langle m\,\big|\left(\sum_{ij} \big|i\big\rangle\beta_{ij}\big\langle j\big|\right)\big|\,n\big\rangle = \sum_{ij} \underbrace{\langle m \,|\, i\rangle}_{=\delta_{mi}}\beta_{ij}\underbrace{\langle j \,|\, n\rangle}_{=\delta_{jn}} = \beta_{mn}.$$

Remember: in Dirac notations a bra-ket $\langle\Phi_1 \,|\, \Phi_2\rangle$ is a number (scalar product), while a ket-bra $\big|\Phi_2\big\rangle\big\langle\Phi_1\big|$ *is an operator*: if it acts on a ket-state $\big|\Psi\big\rangle$ from the left, then the bra $\big\langle\Phi_1\big|$ with the the ket $\big|\Psi\big\rangle$ will produce a number $\alpha = \langle\Phi_1 \,|\, \Psi\rangle$ and the result will be this number times $\big|\Phi_2\big\rangle$:

$$\big|\Psi\big\rangle \longrightarrow \big|\Phi_2\big\rangle\langle\Phi_1 \,|\, \Psi\rangle = \alpha\big|\Phi_2\big\rangle\,. \tag{2.13}$$

[1] a.k.a. the **resolution of identity**

Of course, a sum of such terms is also an operator. Another way to see this is to use the correspondence of kets with column vectors, $\big|\Phi_1\big\rangle \mapsto \{\alpha_n\}^{\mathrm{T}}$ and bras with row vectors, $\big\langle\Phi_2\big| \mapsto \{\beta_n^*\}$ in the same basis. The standard matrix multiplication of column vector (on the left) and row vector (on the right) is a matrix representation of the operator $\big|\Phi_2\big\rangle\big\langle\Phi_1\big|$:

$$\big|\Phi_2\big\rangle\big\langle\Phi_1\big| \mapsto \begin{pmatrix}\alpha_1\\ \alpha_2\\ \vdots\end{pmatrix}\left(\beta_1^*, \beta_2^*, \ldots\right) = \begin{pmatrix}\alpha_1\beta_1^* & \alpha_1\beta_2^* & \ldots\\ \alpha_2\beta_1^* & \alpha_2\beta_2^* & \ldots\\ \vdots & \vdots & \ddots\end{pmatrix}. \tag{2.14}$$

A special case is when we use the ***eigenset*** of operator $\hat{B}$ as a basis, provided that it is complete. The eigenset is a set of eigenvectors, $\big\{\big|\beta_n\big\rangle\big\}$ of $\hat{B}$, that satisfy the eigenvalue equation (EVE):

$$\hat{B}\big|\beta_n\big\rangle = \beta_n\big|\beta_n\big\rangle \tag{2.15}$$

(we use here a more specific notation, $\big|\beta_n\big\rangle$ instead of generic $\big|n\big\rangle$). Completeness means, as usual, that $\sum_n \big|\beta_n\big\rangle\big\langle\beta_n\big| = \hat{I}$. Then we have

$$\hat{B} = \hat{B}\hat{I} = \sum_n \hat{B}\big|\beta_n\big\rangle\big\langle\beta_n\big| = \sum_n \beta_n\big|\beta_n\big\rangle\big\langle\beta_n\big|. \tag{2.16}$$

In particular, from the most famous EVE in physics, the time-independent Schrödinger equation, follows the Dirac representation of the Hamiltonian in its own eigenvalue basis, which is the eigenenergy basis:

$$\hat{H}\big|\varepsilon_n\big\rangle = \varepsilon_n\big|\varepsilon_n\big\rangle \qquad \Rightarrow \qquad \hat{H} = \sum_n \varepsilon_n\big|\varepsilon_n\big\rangle\big\langle\varepsilon_n\big|. \tag{2.17}$$

2.3 Adjoint and self-adjoint (Hermitian) operators

In the previous lecture we introduced quantum mechanical operators acting on vectors in Hilbert space. We have only considered their action on ket-vectors resulting in another ket-vector:

$$\big|\Phi\big\rangle = \hat{O}\big|\Psi\big\rangle = \big|\hat{O}\Psi\big\rangle\,. \tag{2.18}$$

Operators can also act on bra-vectors transforming them into another bra-vector:

$$\big\langle\tilde{\Phi}\big| = \big\langle\Psi\big|\hat{O}\,. \tag{2.19}$$

I have deliberately used tilde in $\big\langle\tilde{\Phi}\big|$ to distinguish the resulting bra-vector from $\big\langle\Phi\big|$ corresponding to the ket-vector $\big|\Phi\big\rangle$. For a general operator $\hat{O}$, bra-vectors $\big\langle\Phi\big|$ and $\big\langle\tilde{\Phi}\big|$ are different. To demonstrate this, consider operator $\hat{O} = \big|X\big\rangle\big\langle Y\big|$ acting on two different vectors $\big|X\big\rangle$ and $\big|Y\big\rangle$. We have

$$\big|\Phi\big\rangle = \big|X\big\rangle\langle Y \mid \Psi\rangle\,, \qquad \big\langle\tilde{\Phi}\big| = \langle\Psi \mid X\rangle\big\langle Y\big|\,. \tag{2.20}$$

The scalar products are just numbers so that $\big|\tilde{\Phi}\big\rangle$ is proportional to $\big|Y\big\rangle$ which is not equal to $\big|\Phi\big\rangle$, proportional to $\big|X\big\rangle$.

If we want to get the bra-vector $\langle\Phi|$ corresponding to $|\Phi\rangle = |\hat{O}\Psi\rangle$ we must act on the bra-vector $\langle\Psi|$ with the *adjoint operator*

$$\langle\Phi| = \langle\Psi|\hat{O}^\dagger = \langle\hat{O}\Psi| \,. \tag{2.21}$$

The "dagger" symbol in $\hat{O}^\dagger$ means Hermitian conjugation of the operator. In most cases, adjoint and Hermitian-conjugate operator means the same.[2]

To make the reason why $\hat{O}^\dagger$ is Hermitian conjugate of $\hat{O}$, let us calculate the matrix elements of $\hat{O}^\dagger$ in a basis $\{|i\rangle\}$, using the skew-symmetry of scalar product, $\langle\Psi \mid \Phi\rangle = \langle\Phi \mid \Psi\rangle^*$:

$$\hat{O}^\dagger_{ij} \equiv \langle i|\hat{O}^\dagger|j\rangle = \langle\hat{O}i \mid j\rangle = \langle j \mid \hat{O}i\rangle^* = \langle j|\hat{O}|i\rangle^* = O^*_{ji} \,, \tag{2.22}$$

This relation shows that the matrix corresponding to the adjoint operator $\hat{O}^\dagger$ is the Hermitian conjugate of the matrix corresponding to $\hat{O}$, *i.e.* is obtained by transposing O_{ij} and taking the complex conjugate of its elements:

$$\hat{O} \longleftrightarrow \begin{bmatrix} O_{11} & O_{12} & \dots \\ O_{21} & O_{22} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} , \qquad \hat{O}^\dagger \longleftrightarrow \begin{bmatrix} O^*_{11} & O^*_{21} & \dots \\ O^*_{12} & O^*_{22} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} . \tag{2.23}$$

The adjoint of a product of two operators is the product of their adjoints in reversed order:

$$(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger \,. \tag{2.24}$$

Let us prove it. Take a ket-vector $|\Psi\rangle$ and act on it with the product $\hat{A}\hat{B}$. The obtained ket-vector $|\hat{A}\hat{B}\Psi\rangle = \hat{A}\hat{B}|\Psi\rangle$ is converted to the corresponding bra-vector and we write

$$\langle\hat{A}\hat{B}\Psi| = \langle\Psi|(\hat{A}\hat{B})^\dagger \,, \tag{2.25}$$

treating $\hat{A}\hat{B}$ as one operator. Now we do this step by step. First we extract $\hat{A}^\dagger$,

$$\langle\hat{A}\hat{B}\Psi| = \langle\hat{B}\Psi|\hat{A}^\dagger \,, \tag{2.26}$$

and use the fact that $\langle\hat{B}\Psi| = \langle\Psi|\hat{B}^\dagger$ so that

$$\langle\hat{A}\hat{B}\Psi| = \langle\Psi|\hat{B}^\dagger\hat{A}^\dagger \tag{2.27}$$

Since Ψ is arbitrary this proves our statement.

The operators which coincide with their adjoints are called **self-adjoint** or **Hermitian**: $\hat{H} = \hat{H}^\dagger$.

They play a major role in quantum mechanics as their eigenvalues are real[3]. Indeed, their *expectation* value (average) in a state $|\Psi\rangle$ is

$$\langle\hat{O}\rangle \equiv \langle\Psi|\hat{O}|\Psi\rangle = \langle\Psi \mid \hat{O}\Psi\rangle = \langle\hat{O}\Psi \mid \Psi\rangle^* = \langle\Psi|\hat{O}^\dagger|\Psi\rangle^* = \langle\Psi|\hat{O}|\Psi\rangle^* \,, \tag{2.28}$$

[2] We will not consider situations when they are not exactly the same.

[3] Similarly we define **anti-Hermitian** operators, $\hat{A}^\dagger = -\hat{A}$, whose eigenvalues happen to be pure imaginary numbers

which is real. From this follows that their *eigenvalues* λ are all real:

$$\hat{O}\big|\lambda_i\big\rangle = \lambda_i\big|\lambda_i\big\rangle \quad \Rightarrow \quad \big\langle\lambda_i\big|\hat{O}\big|\lambda_i\big\rangle = \lambda_i\langle\lambda_i \mid \lambda_i\rangle\,. \tag{2.29}$$

The left-hand side is real as well as the scalar product in the right-hand side. Therefore, all λ_i are real. Hence, they are suitable to represent physically measurable quantities – *observables*. Of course, there are non-Hermitian operators that have only real eigenvalues. However, if we demand that the solutions of the EVE for operator $\hat{O}$ with real eigenvalues are orthonormalised, then its Dirac representation is

$$\hat{O} = \sum_n \lambda_n\big|\lambda_n\big\rangle\big\langle\lambda_n\big| \quad \Rightarrow \quad \hat{O}^\dagger = \sum_n \lambda_n^*\big|\lambda_n\big\rangle\big\langle\lambda_n\big| = \sum_n \lambda_n\big|\lambda_n\big\rangle\big\langle\lambda_n\big| \quad \Rightarrow \quad \hat{O}^\dagger = \hat{O}^\dagger$$

There exist non-Hermitian operators with real eigenvalues. Indeed, making a similarity transformation of $\hat{H}_\mathrm{d} = \mathrm{diag}\,(\lambda_1 \cdots \lambda_n)$, we find $\hat{H} = \hat{P}\hat{H}_\mathrm{d}\hat{P}^{-1}$ to be, in general, non-Hermitian. To see why the Hermitian operators play a special role in QM, we need also to see why the unitary operators are equally important.

2.4 Unitary operators

Another type of operators related to their adjoints are **unitary** operators defined by

$$\hat{U}^\dagger = \hat{U}^{-1} \qquad \text{or} \qquad \hat{U}^\dagger\hat{U} = \hat{U}\hat{U}^\dagger = \hat{I}\,. \tag{2.30}$$

They play a role similar to that of complex numbers of unit modulus, $u = e^{\mathrm{i}\theta}$. These operators preserve scalar product of vectors they act on: let $\big|\Phi\big\rangle_1 = \hat{U}\big|\Psi\big\rangle_1$ and $\big|\Phi\big\rangle_2 = \hat{U}\big|\Psi\big\rangle_2$, then

$$\langle\Phi_1 \mid \Phi_2\rangle = \big\langle\Psi_1\big|\hat{U}^\dagger\hat{U}\big|\Psi_2\big\rangle = \big\langle\Psi_1\big|\hat{I}\big|\Psi_2\big\rangle = \langle\Psi_1 \mid \Psi_2\rangle\,. \tag{2.31}$$

You might remember from Maths that any unitary operator, $\hat{U}$, can be represented as an exponent of some Hermitian operator, $\hat{h}$:

$$\hat{U} \equiv \mathrm{e}^{i\hat{h}} \quad \Rightarrow \quad \hat{U}^\dagger = \mathrm{e}^{-i\hat{h}^\dagger} = \mathrm{e}^{-i\hat{h}} = \hat{U}^{-1}. \tag{2.32}$$

All basis transformations in QM must be unitary to preserve normalisation. If we unitary transform $\hat{H}_\mathrm{d} = \mathrm{diag}\,(\lambda_1 \cdots \lambda_n)$ with real λ_n, we find $\hat{H} = \hat{U}\hat{H}_\mathrm{d}\hat{U}^\dagger$ to be Hermitian:

$$\hat{H}^\dagger = \left(\hat{U}\hat{H}_\mathrm{d}\hat{U}^\dagger\right)^\dagger = \hat{U}\hat{H}_\mathrm{d}^*\hat{U}^\dagger = \hat{H}$$

since $\hat{H}_\mathrm{d}^\dagger = \hat{H}_\mathrm{d}^* = \hat{H}_\mathrm{d}$ as all the eigenvalue are real.

The requirement of all the eigenvalues being real and all the basis transformation being orthogonal leads to the formulation of the next two QM postulates.

2.5 Quantum Measurements and the 2nd and 3rd QM Postulates

QM Postulate II. Any observable is described by a corresponding Hermitian operator; measurable values of this observable are eigenvalues of this operator.

This is the second of the four postulates[4] that lie in the foundation of QM. Naturally, a postulate is not provable but something taken for granted.

QM got its name since measurements of many (albeit not all) microscopic observables return well defined ***quantised*** values. The discreteness of atomic optical spectra, in particular of hydrogen, could not be explained classically and prompted a revolutionary new explanation by Niels Bohr, which is now known as semiclassical. This was great but not sufficient, and entirely new quantum mechanics emerged ten years later in seminal works of Erwin Schrödinger – wave mechanics, and Werner Heisenberg – matrix mechanics. Both these approaches explained known "quantum" experiments equally well but seemed mutually contradictory competing theories until Heisenberg and Dirac proved their full equivalence.

While everybody (qualified to judge) is happy with the 2nd postulate, the next one was and remains to be most controversial part of the QM.

QM Postulate III. A measurement of observable $\hat{A}$ in a QM system in state $\big|\Psi\big\rangle$ yields one of its eigenvalues, A_n, with the probability $P_n = |\langle A_n \,|\, \Psi\rangle|^2$. As a result of the measurement, the state will instantaneously collapse from $\big|\Psi\big\rangle$ into $\big|A_n\big\rangle$.

Here an observable $\hat{A}$ (which could be energy, momentum, position, angular momentum, *etc*) obeys the EVE $\hat{A}\big|A_n\big\rangle = A_n\big|A_n\big\rangle$, while state $\big|\Psi\big\rangle$ is assumed to be normalised and the eigenset $\big\{\big|A_n\big\rangle\big\}$ is orthonormal, $\langle A_n \,|\, A_m\rangle = \delta_{nm}$, and complete, $\sum_n \big|A_n\big\rangle\big\langle A_n\big| = \hat{I}$. This means that any $\big|\Psi\big\rangle$ can be represented as a ***quantum superposition*** of the eigenstates $\big\{\big|A_n\big\rangle\big\}$

$$\big|\Psi\big\rangle = \hat{I}\big|\Psi\big\rangle = \Big(\sum_n \big|A_n\big\rangle\big\langle A_n\big|\Big)\big|\Psi\big\rangle = \sum_n \langle A_n \,|\, \Psi\rangle\big|A_n\big\rangle \tag{2.33}$$

The formulation given here is known as the Copenhagen interpretation of QM (wholeheartedly accepted by the ***qualified*** majority[5]). The collapse of state $\big|\Psi\big\rangle$ means that the measurement of $\hat{A}$ that gives an eigenvalue A_n ***projects*** $\big|\Psi\big\rangle$ into the corresponding eigenstate $\big|A_n\big\rangle$ and implies that any subsequent measurement of $\hat{A}$ performed *immediately after the first one* will result in the same particular value A_n as in the first measurement.[6] Equation (2.33) ensures $\sum_n P_n = \sum_n |\langle A_n \,|\, \Psi\rangle|^2 = 1$, and the probabilities defined in the 3rd Postulate give the *expectation* (average) value of the observable in the standard QM way:

$$\langle A\rangle = \sum_n A_n P_n = \sum_n A_n |\langle A_n \,|\, \Psi\rangle|^2 = \sum_n \langle\Psi \,|\, A_n\rangle A_n \langle A_n \,|\, \Psi\rangle = \big\langle\Psi\big|\hat{A}\big|\Psi\big\rangle. \tag{2.34}$$

In the next lecture, we discuss why this postulate is so controversial and look in more detail on the simplest example of quantum measurements following the 3rd postulate.

[4] In some books, postulates are grouped as 5 or 6 statements, but their contents is the same.

[5] The minority that never accepted this interpretation included Einstein and Schrödinger himself.

[6] The "immediately after" clause is important as the collapsed state may change as a result of evolution which we will discuss soon.

Lecture 3. Two-level systems. Quantum measurements.

In this lecture, we consider consequences of the 2$^{\rm nd}$ and 3$^{\rm rd}$ postulates and start discussing inevitable controversies around the probabilistic nature of Quantum Mechanics.

A full physical interpretation of Quantum Mechanics emerged as a result of intensive discussions between Bohr and Heisenberg and is now known as the ***Copenhagen interpretation***, accepted by the overwhelming majority of physicists.[1] The focus of this interpretation is the intrinsically probabilistic character of QM measurements. When more than one outcome of a measurement may occur, it is impossible to predict it with certainty. According to the QMP3, only the probability of the outcome could be given (equal to $|\langle\Psi \mid A_n\rangle|^2$ for measuring the eigenvalue A_n of $\hat{A}$ in state $\big|\Psi\big\rangle$) – that is the fraction of its appearance after repeating the measurement multiple times.

A random result is unpleasant from classical viewpoint (*God does not play dice* according to Einstein) and controversy related to such a statement was made quite apparent by Schrödinger, offering the eponymous cat as a counter-example. The *Schrödinger's cat* appeared as a reaction to the equally famous objection to QM (or to its Copenhagen interpretation) in the paper by Einstein, Podolsky, and Rosen which became known as the *EPR "paradox"*. Before going to consider these objections and their resolution, let us consider in more detail how the QM postulates work using the simplest possible QM system.

3.1 Two-level system.

The two-level system (TLS), which is now much better known under the name *qubit*, is generated by an operator, say $\hat{\sigma}$, that has only two non-degenerate eigenvalues, equal ± 1 in proper units (or spin up-down for a real spin projection equal $\pm\frac{1}{2}\hbar$ or any "pseudo-spin" which is a possible way of mapping a TLS onto). The corresponding Hilbert space is two-dimensional, and the EVE for $\hat{\sigma}$ and its Dirac representation can be written as

$$\begin{aligned} &\hat{\sigma}\big|\sigma\big\rangle = \sigma\big|\sigma\big\rangle, \quad \text{with } \sigma = \pm 1 \quad \Longleftrightarrow \quad \hat{\sigma}\big|\pm\big\rangle = \pm\big|\pm\big\rangle \\ \Rightarrow \quad &\hat{\sigma} = \sum_{\sigma=\pm 1} |\sigma\rangle\sigma\langle\sigma| = |+\rangle\langle+| - |-\rangle\langle-| \equiv |\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow| \equiv |1\rangle\langle 1| - |0\rangle\langle 0|. \end{aligned} \tag{3.1}$$

Here we use the flexibility of Dirac's notations: two eigenstates could be denoted in an arbitrary manner, so that we have chosen $\big|1\big\rangle \equiv \big|\uparrow\big\rangle \equiv \big|+\big\rangle$ and $\big|0\big\rangle \equiv \big|-\big\rangle \equiv \big|\downarrow\big\rangle$.

A spin $\frac{1}{2}$ particle in a magnetic field (traditionally, along the z-axis) is the simplest realisation of TLS , about which you have learned a bit in QM2. This justifies notations: $\big|+\big\rangle \equiv \big|\uparrow\big\rangle$ or $\big|-\big\rangle \equiv \big|\downarrow\big\rangle$ are the states with the spin projection up or down, *i.e.* in the positive or negative direction along the

[1] There is no democracy in science, and a truthfulness – or otherwise – of a physical theory is not decided by the majority vote. Objections of a *qualified* minority (rather than people in the street) could be very important and relevant. The whole modern building of QM has emerged notwithstanding (but rather due to) the objections of the minority that never accepted the Copenhagen interpretation. This minority included, amongst others, Einstein, Schrödinger, Louis de Broglie – all are the Nobel Prize winners with tremendous contributions to QM.

z-axis. The dimensionality of spin = that of the angular momentum is $[p \times r]$ which is the same as that of $\hbar$, and the spin-projection operator $\hat{S}_z$ is related to $\hat{\sigma}$ by $\hat{S}_z = \dfrac{\hbar}{2}\hat{\sigma}$.

The matrix representation of the operator $\hat{\sigma}$ in the 2-dimensional Hilbert space is given by

$$\langle \sigma \,|\, \hat{\sigma} \,|\, \sigma' \rangle = \begin{pmatrix} \langle\uparrow| \left(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow| \right) |\uparrow\rangle & \langle\uparrow| \left(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow| \right) |\downarrow\rangle \\ \langle\downarrow| \left(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow| \right) |\uparrow\rangle & \langle\downarrow| \left(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow| \right) |\downarrow\rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{3.2}$$

where you should recognise (from QM2 and/or Maths2B) Pauli's σ_z matrix. The simplification of the matrix element are based on the orthonormality, *i.e.* $\langle\uparrow|\downarrow\rangle = \langle\downarrow|\uparrow\rangle = 0$ and $\langle\uparrow|\uparrow\rangle = \langle\downarrow|\downarrow\rangle = 1$. Its eigenstates, $|\sigma\rangle = |\pm\rangle$, in matrix representation are

$$|+\rangle \equiv |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \text{and} \qquad |-\rangle \equiv |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.3}$$

Any state of a TLS (qubit, spin $\frac{1}{2}$, *etc*) is a quantum superposition of its eigenstates, $|\Psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$. The requirement for $|\Psi\rangle$ to be normalised gives

$$1 = \langle \Psi \,|\, \Psi \rangle = |\alpha|^2 \underbrace{\langle\uparrow|\uparrow\rangle}_{=1} + |\beta|^2 \underbrace{\langle\downarrow|\downarrow\rangle}_{=1} + \alpha^*\beta \underbrace{\langle\uparrow|\downarrow\rangle}_{=0} + \alpha\beta^* \underbrace{\langle\downarrow|\uparrow\rangle}_{=0} = |\alpha|^2 + |\beta|^2 ,$$

from which follows that α and β could be chosen as sine and cosine of some angle with an arbitrary phase factor. We shall learn (much later in the course) that for spin $\frac{1}{2}$ this angle must be chosen as $\theta/2$ when projections measured along a straight line at angle θ to the z-axis,[2] so that the quantum superposition $\alpha|\uparrow\rangle + \beta|\downarrow\rangle$ could be written as

$$|\Psi\rangle = \mathrm{e}^{i\phi} \cos(\theta/2)|\uparrow\rangle + \mathrm{e}^{i\varphi} \sin(\theta/2)|\downarrow\rangle , \tag{3.4}$$

where phase angles ϕ and φ are arbitrary. According to the 3rd QMP, a measurement can only return

$$|\uparrow\rangle \text{ with the probability } \cos^2(\theta/2) \qquad \text{or} \qquad |\downarrow\rangle, \text{ with the probability } \sin^2(\theta/2) \tag{3.5}$$

while the particle is in the superposition (3.4), *i.e.* simultaneously in states $|\uparrow\rangle$ and $|\downarrow\rangle$ prior to the measurement. Naturally, there is nothing special about the z-axis, and the same probabilities are for the measurement at a given angle θ to any direction.

Finally, the Hamiltonian of any TLS can be written in its own (energy) basis as

$$\hat{H} = \sum_{n=0,1} \varepsilon_n |n\rangle\langle n| = \varepsilon_0 |0\rangle\langle 0| + \varepsilon_1 |1\rangle\langle 1|.$$

The standard EVEs immediately follow from this notation. E.g., the EVE for ε_0:

$$\hat{H}|0\rangle = \varepsilon_0 |0\rangle\langle 0| \,|0\rangle + \varepsilon_1 |1\rangle\langle 1| \,|0\rangle \equiv \varepsilon_0 \,|0\rangle \underbrace{\langle 0|\,0\rangle}_{=1} + \varepsilon_1 \,|0\rangle \underbrace{\langle 1\,|\,0\rangle}_{=0} = \varepsilon_0 \,|0\rangle \ .$$

[2] If we measure an orbital angular momentum, the angle in superposition (3.4) would be θ: the reason for an additional $\frac{1}{2}$ is that spin is not transformed as a vector; one need a rotation through 4π rather than 2π to return to the same spin state, as we shall see later in the course.

Example 3.1. *The TLS is described by the Hamiltonian* $\hat{H} = -\varepsilon\big|0\big\rangle\big\langle 0\big| + \varepsilon\big|1\big\rangle\big\langle 1\big|$. *It is acted upon by an operator* $\hat{A}$ *which has eigenvalues* $A_{1,2} = \pm 1$ *with eigenvectors* $\big|A_{1,2}\big\rangle$ *related to those of* $\hat{H}$ *by*

$$\big|A_1\big\rangle = \tfrac{1}{2}\big|0\big\rangle + \tfrac{\sqrt{3}}{2}\big|1\big\rangle \text{ and } \big|A_2\big\rangle = \tfrac{\sqrt{3}}{2}\big|0\big\rangle - \tfrac{1}{2}\big|1\big\rangle \ .$$

(i) If the system was prepared in the groundstate, what is the probability to measure A_1*?*
(ii) What is the energy of the system after it was measured in state $\big|A_1\big\rangle$*?*
(iii) Write $\hat{A}$ *in the energy basis.*

According to the QMP3,

$$P_1 = |\langle A_1 \,|\, 0\rangle|^2 = \Big|\Big(\tfrac{1}{2}\langle 0| + \tfrac{\sqrt{3}}{2}\langle 1|\Big)|0\rangle\Big|^2 = \Big|\tfrac{1}{2}\underbrace{\langle 0 \,|\, 0\rangle}_{=1} + \tfrac{\sqrt{3}}{2}\underbrace{\langle 1 \,|\, 0\rangle}_{=0}\Big|^2 = \tfrac{1}{4}\,.$$

The energy in state $\big|\Psi\big\rangle$ *is* $\big\langle \Psi \,\big|\, \hat{H} \,\big|\, \Psi\big\rangle$. *In state* $\big|A_1\big\rangle$ *we find*

$$\begin{aligned}\big\langle \hat{H}\big\rangle = \big\langle A_1 \,\big|\, \hat{H} \,\big|\, A_1\big\rangle &= \Big(\tfrac{1}{2}\langle 0| + \tfrac{\sqrt{3}}{2}\langle 1|\Big)\hat{H}\Big(\tfrac{1}{2}|0\rangle + \tfrac{\sqrt{3}}{2}|1\rangle\Big) \\ &= \Big(\tfrac{1}{2}\langle 0| + \tfrac{\sqrt{3}}{2}\langle 1|\Big)\Big(-\tfrac{\varepsilon}{2}|0\rangle + \tfrac{\sqrt{3}\varepsilon}{2}|1\rangle\Big) = -\tfrac{\varepsilon}{4}\langle 0|\,0\rangle + \tfrac{3\varepsilon}{4}\langle 1|\,1\rangle = \tfrac{1}{2}\varepsilon\,.\end{aligned}$$

Expanding $\hat{A}$ *in the energy basis gives (using* $\hat{A}\big|A_n\big\rangle = A_n\big|A_n\big\rangle$*):*

$$\begin{aligned}\hat{A} &= A_1\big|A_1\big\rangle\big\langle A_1\big| + A_2\big|A_2\big\rangle\big\langle A_2\big| = \big|A_1\big\rangle\big\langle A_1\big| - \big|A_2\big\rangle\big\langle A_2\big| \\ &= \Big(\tfrac{1}{2}|0\rangle + \tfrac{\sqrt{3}}{2}|1\rangle\Big)\Big(\tfrac{1}{2}\langle 0| + \tfrac{\sqrt{3}}{2}\langle 1|\Big) - \Big(\tfrac{\sqrt{3}}{2}|0\rangle - \tfrac{1}{2}|1\rangle\Big)\Big(\tfrac{\sqrt{3}}{2}\langle 0| - \tfrac{1}{2}\langle 1|\Big) \\ &= -\tfrac{1}{2}|0\rangle\langle 0| + \tfrac{\sqrt{3}}{2}|0\rangle\langle 0|1 + \tfrac{\sqrt{3}}{2}|1\rangle\langle 1|0 + \tfrac{1}{2}|1\rangle\langle 1|\,.\end{aligned}$$

In matrix form,

$$\mathsf{A} = \frac{1}{2}\begin{pmatrix} -1 & \sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix} .$$

3.2 Controversies around the 3rd postulate

The measurement theory was codified by John von Neumann who clarified that the result of a *quantum* measurement can only be read by a *classical* device, *i.e.* a state $\big|\Psi\big\rangle$ collapses after the measurement as a result of the interaction of a quantum system with such a device. Hence, QM is incomplete by construction as a measurement (collapse) cannot be described by a QM equation.

This incompleteness was one of the main reasons for mighty criticism of QM by many leading physicists focused on unacceptability of the 3$^{\rm rd}$ postulate. The 3$^{\rm rd}$ postulate contains two statements, each being impossible in classical physics. First, the result of a quantum measurement is random: experimentalists know the probabilities of possible outcomes, but cannot predict which particular one will turn up after any particular measurement. So it goes against the following well-known statement: "*Insanity is doing the same thing, over and over again, but expecting different results*", which is often *misattributed* to Einstein,[3] who actually said "*God does not play dice with the Universe*".

[3]One could think that a classical throwing a coin also leads to an unpredictable result but this is only due to the fact

The second controversial statement – an instantaneous collapse of the wavefunction – apart from making QM incomplete seems, on the face of it, to contradict the Relativity Theory, as had been noticed in an extremely important, famous paper by Einstein, Podolsky and Rosen. Their statement subsequently became known as the EPR paradox, which we consider in the following lecture.

Some most spectacular objections to the 3$^{\rm rd}$ QM Postulate have been inspired by (even if not all directly connected to) the EPR paper, focusing on the *conjecture* of the ***hidden variables*** which would make QM complete. In this approach, QM is incomplete by our lack of the proper description, not by nature. The idea is that the measurement outcomes which only appear random but “in reality" are governed by these hidden variables. We postpone considerations of this conjecture (and its spectacular experimental failure) till the following lecture.

Here I start with briefly analyzing the famous ***Schrödinger's cat*** – an objection against the probabilistic nature based on the re-interpretation of the TLS results.

3.3 Schrödinger's cat.

Schrödinger, absolutely unhappy with the probabilistic interpretation of QM, has suggested the following *gedanken* experiment: consider a radioactive atom as a TLS. Over the time it might decay but the timing is unpredictable.[4] Hence, prior to the measurement the atom is in a quantum superposition of (3.4), where $\big|\uparrow\big\rangle$ is the undecayed atom and $\big|\downarrow\big\rangle$ is the decayed one. Further assume that the decay triggers an electronic device that opens an ampule with poison, and put this assembly into a non-transparent, isolated cage with a cat inside.[5] According to the 3$^{\rm rd}$ QMP, we do not know the state before the measurement (opening the door of the cage), so that not only the atom but the cat itself is in a quantum superposition. Hence we can now identify our states as

$$\big|\downarrow\big\rangle - \text{dead}, \qquad\qquad \big|\uparrow\big\rangle - \text{alive}. \tag{3.6}$$

We can define the vitality operator (related to $\hat{\sigma}$ by a shift of eigenvalues)

$$\hat{V}\big|\uparrow\big\rangle = 1\big|\uparrow\big\rangle \text{ alive} \qquad\qquad \hat{V}\big|\downarrow\big\rangle = 0\big|\downarrow\big\rangle \text{ dead} \tag{3.7}$$

Before making a classical measurement of the vitality (opening the door), we do not know whether the atom decayed or not. The measurement of $\hat{V}$ would resolve the issue and give the eigenvalue 1 (alive!) with probability $P = \cos^2(\theta/2)$ and 0 (dead) with probability $\sin^2(\theta/2)$. Here P is averaged over many experiments vitality of the cat[6], while before any particular measurement the cat is in a quantum superposition state of being dead and alive simultaneously.

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that, in general, it is impossible to exactly reproduce initial conditions, in contrast to some experiments in the microworld where the reproducibility of the initial conditions is not a problem but the result of a measurement is unpredictable.

[4] This is by itself classically unpleasant: even when everything is known about the current state of, say, an uranium atom, QM states that it is in principle impossible to predict when it decays. But QM predicts with an arbitrary accuracy the average half-decay time, i.e. the time over which half of the atoms in any given lump of uranium decay.

[5] RSPA would prosecute those who try to make such an experiment in reality.

[6] We use the same notations as for the spin; here θ by itself has no physical meaning.

Obviously, such a conclusion is very unpleasant as we believe (with a good reason to be spelt out in subsequent lectures) that QM only describes microscopic objects.[7] This remains the subject of (mostly) philosophical disputes even now.

However, for the majority of researchers the following Bohr's answer to Schrödinger was persuasive: the measurement is an interaction of a quantum object with a classical measuring device, whether this device is observed by experimentalists or not. From this viewpoint, triggering the classical electronic device that opens the ampule is already a measurement, even if not witnessed by a human being. Hence, the poor cat is certainly dead if the device was triggered and alive if it was not, which happens (or not) independently of opening the door.

Nevertheless, the Schrödinger cat became proverbial, known in popular literature, and used by physicists for describing most entangled states (see the next lecture) as *cat states*.

Exercise: write down a state for a Schrödinger's cat who is four times more likely to be alive than dead and calculate its vitality.

[7] There are exceptions like superfluidity or superconductivity in macroscopic systems which are quantum phenomena that cannot be explained classically; however, the Schrödinger cat does not belong to this category.

Lecture 4. Entanglement and (the absence of) hidden variables

Here we briefly consider the famous EPR paradox, introduce the concept of quantum entanglement and discuss the Bell's inequality as the criterion of existence (or otherwise) of hidden variables that would allow one to get rid of quantum randomness.

The Schrödinger cat, discussed in the previous lecture, has emerged as Schrödinger's reaction to the famous paper by Einstein, Podolsky and Rosen published in 1935, which questioned the completeness of QM. The conclusion of the paper became known as the EPR paradox, and we shall consider it here in a simplified (but equivalent) form suggested by Bohm.

4.1 EPR paradox

The Schrödinger cat was a macroscopic extension of a TLS exemplified by spin $\frac{1}{2}$. Now we consider the system that contains two identical spin $\frac{1}{2}$ particles, created by the decay of a 0 spin particle at rest in the laboratory frame. The momentum conservation means that these two particles are moving in the opposite directions at the same speed, while the angular momentum conservation means that they have the opposite spin projections.

From QM2, you know that the spin projection can be measured using the Stern-Gerlach (SG) magnet oriented along any axis. Let us assume that Alice and Bob measure these projections for particles A and B, with Bob being slightly farther away from the point of decay so that Alice makes her measurement slightly earlier. Let us further assume that they are separated by a spacelike interval so that Alice cannot communicate the result of her measurement to Bob before his measurement, even if she sends a signal at the speed of light.

Suppose that Alice and Bob have oriented their SG magnets along the z axis. Their measurements can return, according to the 3$^{\rm rd}$ QMP, only the eigenvalues of the spin-projection operator, *i.e.* $\pm\frac{\hbar}{2}$. Then if Alice measured $s_z = +\frac{\hbar}{2}$, Bob ***must*** measure $s_z = -\frac{\hbar}{2}$. Thus their measurements are ***correlated***, in spite of the fact that they could not exchange any information.

This is the essence of the EPR paradox: the particle B somehow 'knows' in what state was the particle A, although any information exchange is forbidden by the theory of relativity. So there is a correlation between A and B, regardless of how far apart are the particles at the time of the measurements. Such a correlation is now known as ***quantum entanglement***.

On the face of it, this implies action at a distance or transmission of information at a superluminal speed, unless any spin component of each particle has *a definite value* from the moment of decay, and the measurements simply reveal these ***predetermined*** values – obviously, in contradiction to the 3$^{\rm rd}$ QM Postulate. For Einstein and collaborators the existence of such objective properties independently from subsequent measurements seemed to be much more reasonable than throwing QM dice, not even speaking about the information exchange at a superluminal speed. Any theory that obey these requirements – measurements at remote points are not correlated as the real-world particles must have objective properties unrelated to measurements – would be called a ***realistic local theory.*** Obviously, the QM which is based, in particular, on the 3$^{\rm rd}$ postulate is neither local, as a state of the

particle depends on a state of the other particle after the measurement, nor 'realistic' as it remains undetermined prior to the measurement.

Bohr, however, has noticed that entanglement was in no contradiction with relativity: although the wavefunction instantaneously collapses on the first (Alice's) measurement and the result of the second (Bob's) measurement is thus predetermined, there is no way to let Bob know prior to his measurement about Alice's result (or even whether Alice has already done the measurement). While accepted by many, this was not persuasive for Einstein and Schrödinger, not mentioning others.

This disagreement led to a very animated debate (lasted for several years) between great and famous physicists[1] that held opposite views on the nature or very existence of entanglement. These views finessed in the debate can be summarised as follows:

1. QM is incomplete due to lack of our knowledge: Bob's and Alice's results are not random as both particles' states are fully determined by some local (inherent to each particle) hidden variables; hence no information is transferred as A and B measured predetermined states

2. QM is nonlocal (and incomplete by nature): The probabilistic description is as good as it gets: results of Bob's and Alice's measurements are not predetermined but mutually *entangled* despite the absence of information transfer between them

For almost thirty years since the start of this famous discussion (often quoted as Einstein vs Bohr, although quite a few almost equally famous physicists participated), it had no definite resolution, and gradually became a subject of philosophical rather than physical discussions, since most physicists were quite satisfied with the fact that QM works, whether there are hidden variables or not. But this has changed after publication in 1964 of work by John Stewart Bell.

4.2 Bell's inequality

After the splash caused by the EPR paper, the interest in foundational matters gradually faded: although a few models of hidden variables were suggested, none was experimentally verifiable (or falsifiable). The vague idea of eventually finding the 'correct' ones was certainly out of the focus but still viable. Then Bell has discovered that no particular model of hidden variables was needed: if they exist, then the measurements results must obey a certain inequality.

Namely, Bell has noticed in 1964 that a mere assumption (inherent to any realistic local theory) that the values of spin projections in any direction were predetermined[2] would lead to a certain verifiable (or falsifiable) analytic inequality *without specifying hidden variables*. The importance of this has been immediately realised by the physics community, and subsequently different Bell-like inequalities have emerged, some of them easier for the experiment than the original one. However, the proof of the original inequality is rather straightforward and we go along with it.

[1] Some quotes: Einstein to Schrödinger – "The talmudic philosopher (Bohr) doesn't give a hoot for 'reality' . . . ", Ehrenfest to Einstein – "Einstein, I am ashamed of you. . . "

[2] You have learned in QM2 that projections of angular momentum or spin in different directions cannot be simultaneously measured as the appropriate operators do not commute. The *hidden variables* conjecture does not assume that any two components can be measured together, but only that any of them is predetermined.

Note first that Alice's and Bob's SG magnets could be arbitrarily aligned with an angle θ between their orientations, say $\boldsymbol{n}_A$ and $\boldsymbol{n}_B$. We have seen that measuring the spin projection of a particle prepared in state $\big|\uparrow\big\rangle$ along the z-axis in the field directed at an angle θ would give $\big|\uparrow\big\rangle$ along the new direction with the probability $\cos^2(\theta/2)$. Now it was Alice's measurement that prepared the spin projection for Bob (in the opposite to her direction): if her measurement produced $s_{\boldsymbol{n}_A} = \frac{\hbar}{2}$, Bob's measurement would give $s_{\boldsymbol{n}_B} = \frac{\hbar}{2}$ with the probability $\cos^2(\theta/2)$. Hence, their measurements would be correlated unless $\theta = \pi/2$ (then Bob's measurements would give $\uparrow$ or $\downarrow$ with probability $\frac{1}{2}$).

Assume that Alice and Bob have agreed, prior to the experiment, to orient their SG magnets in ***any one*** of three ***coplanar*** directions, $\boldsymbol{n}_1$, $\boldsymbol{n}_2$ and $\boldsymbol{n}_3$. If local hidden variables exist, then we can denote particle A state as $\big|\sigma_1, \sigma_2, \sigma_3\big\rangle$, where all the three components are predetermined with each σ_i being either up or down[3], *e.g.*, $\big|\uparrow, \downarrow, \uparrow\big\rangle$. Similarly predetermined is the state of particle B, $\big|\overline{\sigma}_1, \overline{\sigma}_2, \overline{\sigma}_3\big\rangle$, where $\big|\overline{\uparrow}\big\rangle = \big|\downarrow\big\rangle$ to allow for fact that the total spin projection must be zero in any direction. Let denote the joined state of two particles as

$$\big|\sigma_1, \sigma_2, \sigma_3;\ \overline{\sigma}_1, \overline{\sigma}_2, \overline{\sigma}_3\big\rangle \equiv \big|\sigma_1, \sigma_2, \sigma_3\big\rangle\,. \tag{4.1}$$

Here we do not need explicitly write spin directions in any outcome of Bob's measurements as they are always opposite to Alice's. Let $f(\sigma_1, \sigma_2, \sigma_3)$ be the fraction of pairs in the above state for any particular values of σ_i for particle A (and corresponding $\overline{\sigma}_i$ for particle B, which we do not explicitly write). It is vital to stress that this correspondence is *predetermined* and measurements are totally local, since the outcome of Alice's measurement does not depend on the orientation of Bob's SG magnets.

Now we compute $F_{\uparrow\uparrow}(\boldsymbol{n}_i; \boldsymbol{n}_j)$, the fraction of outcomes where Alice's measurement along $\boldsymbol{n}_i$ and Bob's measurement along $\boldsymbol{n}_j$ give the up-projection of spin. Clearly, $i \neq j$ here as the projections along the same axis must be opposite. The spin projection conservation also fixes outcomes along these two directions for both A and B: Alice's expected spin projection for $\boldsymbol{n}_j$ is always $\downarrow$ as it is opposite to Bob's. Hence to get F one needs to sum over two possible outcomes in the remaining direction. Then, choosing three (out of the six possible) pairs of directions, we find

$$\begin{aligned}
F_{\uparrow\uparrow}(\boldsymbol{n}_1; \boldsymbol{n}_2) &= \sum_{\sigma_3} f(\uparrow, \downarrow, \sigma_3) = f(\uparrow, \downarrow, \uparrow) + f(\uparrow, \downarrow, \downarrow)\,; \\
F_{\uparrow\uparrow}(\boldsymbol{n}_1; \boldsymbol{n}_3) &= \sum_{\sigma_2} f(\uparrow, \sigma_2\, \downarrow) = f(\uparrow, \uparrow, \downarrow) + f(\uparrow, \downarrow, \downarrow)\,; \\
F_{\uparrow\uparrow}(\boldsymbol{n}_3; \boldsymbol{n}_2) &= \sum_{\sigma_1} f(\sigma_1, \downarrow, \uparrow) = f(\uparrow, \downarrow, \uparrow) + f(\downarrow, \downarrow, \uparrow)\,.
\end{aligned} \tag{4.2}$$

From this we obtain

$$\boxed{F_{\uparrow\uparrow}(\boldsymbol{n}_1; \boldsymbol{n}_3) + F_{\uparrow\uparrow}(\boldsymbol{n}_3; \boldsymbol{n}_2) - F_{\uparrow\uparrow}(\boldsymbol{n}_1; \boldsymbol{n}_2)} = f(\uparrow, \uparrow, \downarrow) + f(\downarrow, \downarrow, \uparrow) \geqslant 0 \tag{4.3}$$

since all the fractions f are non-negative. This is one of the famous Bell's inequalities.

[3]The 2nd QM postulate claim that only eigenvalues of operators can be measured is not disputed in any formulation.

Let me stress again: for any *realistic local theory* this inequality (and a plethora of similar ones) must be obeyed, whatever is the nature of local hidden variables. From this viewpoint, it is (almost) immediately clear that QM in the Copenhagen interpretation is not one of such theories. To see this, let us calculate the probabilities $P_{\uparrow\uparrow}(\boldsymbol{n}_i;\boldsymbol{n}_j) = P_{\uparrow\uparrow}(\theta_{ij})$, which are the QM counterparts of frequencies $F_{\uparrow\uparrow}(\boldsymbol{n}_i,\, \boldsymbol{n}_j)$ of (4.2). Here θ_{ij} is the angle between Alice's and Bob's SG magnets. It is straightforwardly based on the results of Section 3.1. Alice has the probability $1/2$ to get spin projection up along $\boldsymbol{n}_i$. Then for Bob's measurement particle B would have spin projection down along $\boldsymbol{n}_i$ with probability 1, and according to (3.5) he will measure $\left|\downarrow\right\rangle$ (*i.e.* the same projection in which particle has been prepared) with the probability $\sin^2(\theta_{ij}/2)$. Hence the QM counterpart of Bell's inequality (4.3) is

$$P_{\uparrow\uparrow}(\boldsymbol{n}_1;\boldsymbol{n}_3) + P_{\uparrow\uparrow}(\boldsymbol{n}_3;\boldsymbol{n}_2) - P_{\uparrow\uparrow}(\boldsymbol{n}_1;\boldsymbol{n}_2) = \sin^2(\theta_{13}/2) + \sin^2(\theta_{23}/2) - \sin^2(\theta_{12}/2) \geqslant 0 \quad (4.4)$$

Now it is trivial to find geometry in which this inequality clearly fails. Choosing, e.g., that $\boldsymbol{n}_3$ bisects the angle between $\boldsymbol{n}_1$ and $\boldsymbol{n}_2$, *i.e.* $\theta_{13} = \theta_{23} = \frac{1}{2}\theta_{12}$ reduces Bell's inequality to

$$2\sin^2(\theta_{13}/2) - \sin^2(\theta_{13}) = 1 - \cos(\theta_{13}) - \sin^2(\theta_{13}) \geqslant 0 \quad \Longleftrightarrow \quad \cos^2(\theta_{13}) \geqslant \cos(\theta_{13}).$$

This inequality breaks down when $\cos(\theta_{13}) \geqslant 0$, *i.e.* for $0 \leqslant \theta_{13} \leqslant \pi/2$.

Hence, the conclusion is that a (relatively) simple experiment could decisively choose between locality and incompleteness on the one hand and QM entanglement on the other.

4.3 Experimental verdict: no hidden variables

Richard Feynman: "It seems to be almost ridiculous that you can squeeze the difficulty of QM to a numerical question that one thing is bigger than another. But there you are – it is bigger..."

The inequality (4.3) was not easy (but still possible) to verify experimentally, but many other Bell-type inequalities have been developed. The first experiment was carried out in 1972 by John Clauser[4] and Stuart Freedman.[5] It confirmed the violation of one of Bell's inequalities (in form suggested by Clauser and Shimony and modified by Freedman), albeit not with a decisive accuracy.

A real breakthrough was due to experiments led by Alain Aspect[4] in 1982 (in collaboration with other three physicists), based on his laying out a strategy for such an experiment in his PhD in 1978. They were based on the most popular CHSH[6] form of Bell's inequalities. The experiment was done with photons that can have only two polarisations, so that they are in this aspect TLS systems. Instead of SG magnets, linear polarisers have been used, each with two possible orientations $\boldsymbol{n}$ and $\boldsymbol{n}'$. The role of spin up and down is played by horizontal (H), $\perp$ to $\boldsymbol{n}$, or vertical (V), $\parallel$ to $\boldsymbol{n}$, polarisation. The CHSH inequality is

$$|S| \leqslant 2 \text{ where } S = E(\boldsymbol{n}_A,\boldsymbol{n}_B) - E(\boldsymbol{n}_A,\boldsymbol{n}'_B) + E(\boldsymbol{n}'_A,\boldsymbol{n}_B) + E(\boldsymbol{n}'_A,\boldsymbol{n}'_B), \quad (4.5)$$

[4] 2022 Nobel Prize in physics awarded to A Aspect, JF Clauser and A Zeilinger "for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science", 04 October 2022

[5] Passed away in 2012 – to get the Nobel Prize one should be clever, lucky and live long enough. Sadly, John Bell has passed away in 1990 but not before he had seen numerous experimental falsifications of his theorem. Now, *John Stewart Bell Prize* is awarded biannually for major advances in foundations of quantum mechanics.

[6] J Clauser, M Horne, A Shimony and R Holt

and E is the correlation between Alice's and Bob's measurements of polarisations,

$$E(A,B) = P_{VV} + P_{HH} - P_{VH} - P_{HV}$$

with, e.g., P_{VV} is the probability (or fraction) of measuring two vertical polarisation along the corresponding axes.

In contrast to this particular form of Bell's inequality, QM allows $|S|$ to reach $2\sqrt{2}$. The first experiments in early 80's produced $|S|$ around 2.8 with fidelity exceeding 5σ. The current value of $|S|$ is 2.8284 with fidelity exceeding 99σ, *i.e.* the probability that this result is wrong is less than $e^{-99} \approx 10^{-42}$. The argument of ***entanglement*** vs ***locality*** has been won by entanglement with knock out.

We will come back to learning more about entanglement (but, sadly, not about its applications – this would require a separate set of lectures) closer to the end of this course.

Lecture 5. Position and Momentum Operators

We shall return to entangled states later, after learning here how to deal with continuum. We introduce the position and momentum operators and formally show that the wavefunction $\Psi(x)$ are components of Dirac state $\big|\Psi\big\rangle$ in the continuous position basis. Then we find the representation of the momentum operator in this basis.

5.1 Continuous basis

Position and momentum are very important in quantum mechanics as they provide a way to write down Hamiltonians by simply replacing classical variables with the corresponding operators. To treat this properly, we should finally consider an infinite-dimensional Hilbert space after having dealt with the TLS - two discrete degrees of freedom correspond to two levels only and thus generate two-dimensional Hilbert space (not to be confused with two-dimensional real space where QM particles with two continuous degrees of freedom generate infinitely-dimensional Hilbert spaces).

We start with a particle in one dimension (the 3D generalisation will be given at the end of this lecture). In quantum mechanics the position x of the particle becomes an observable and is represented by the *position operator* $\hat{x}$. Similarly to any Hermitian operator the position operator defines its eigenbasis (set of eigenvectors):

$$\hat{x}\big|x\big\rangle = x\big|x\big\rangle\,. \tag{5.1}$$

This notation is a bit heavy, but it is pretty much standard in modern literature. In the equation above $\hat{x}$ is an operator, *any* real number x is its eigenvalue and $\big|x\big\rangle$ is the corresponding eigenvector. In the previous lectures we have assumed eigenvalues of an operator form a discrete, countable set, allowing us to use notations like $\big\{\big|n\big\rangle\big\}$ (or $\big\{\big|\beta_n\big\rangle\big\}$, if it were necessary to be more specific). On the contrary, the eigenvalues of the position operator are all real numbers (a QM particle can be anywhere) forming a *continuous set*. This leads to some important modifications. A summation over discrete eigenvalues goes over to an integration, so the **completeness** and **orthonormality** conditions in the case of the position eigenbasis becomes

$$\hat{I} = \int_{-\infty}^{\infty} \mathrm{d}x\,\big|x\big\rangle\big\langle x\big| \quad \text{and} \quad \big\langle x\,\big|\,x'\big\rangle = \delta(x-x')\,. \tag{5.2}$$

Here $\delta(x)$ is the Dirac delta function defined by the relation

$$\int_{-\infty}^{\infty} \mathrm{d}x' f(x')\delta(x-x') = f(x)\,. \tag{5.3}$$

Similar to the diagonal matrix representation of $\hat{B}$ in its eigenbasis, $\big\langle n\,\big|\,\hat{B}\,\big|\,m\big\rangle = \beta_n\delta_{nm}$, we have

$$\big\langle x'\,\big|\,\hat{x}\,\big|\,x\big\rangle = x\langle x'\,|\,x\rangle = x\delta(x-x') \quad \Rightarrow \quad \big\langle x'\,\big|\,\hat{V}(x)\,\big|\,x\big\rangle = V(x)\delta(x-x'). \tag{5.4}$$

where $\hat{V}(x) \equiv V(\hat{x})$ is any operator that depends on x only, *e.g.*, a potential energy operator, and $V(x)$ is simply the corresponding function of x, assumed to be smooth, so that the last equality above could be verified by Taylor's expansion.

Now we can represent a state $\big|\Psi\big\rangle$ of a particle in 1D using the orthonormal and complete position eigenbasis $\big\{\big|x\big\rangle\big\}$ similarly to representation with a discrete basis $\big\{\big|n\big\rangle\big\}$, using the resolution of unity:

$$\big|\Psi\big\rangle = \hat{I}\big|\Psi\big\rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, \big|x\big\rangle\langle x \,|\, \Psi\rangle \equiv \int_{-\infty}^{\infty} \mathrm{d}x\, \Psi(x)\big|x\big\rangle \tag{5.5}$$

where we have identified the component $\langle x \,|\, \Psi\rangle$ of state vector Ψ in the position basis with the familiar **wavefunction** $\Psi(x)$. The skew symmetry of the inner product gives us its complex conjugate counterpart $\langle \Psi \,|\, x\rangle = \langle x \,|\, \Psi\rangle^* = \Psi^*(x)$. This correspondence produces the **inner product** of two state vectors in terms of the integral of wavefunctions as expected:

$$\langle \Psi \,|\, \Phi\rangle = \big\langle\Psi\big|\hat{I}\big|\Phi\big\rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, \langle \Psi \,|\, x\rangle\langle x \,|\, \Phi\rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, \Psi^*(x)\Phi(x)\,. \tag{5.6}$$

Inserting $\hat{I}$ twice into the Dirac expression for the average of an x dependent observable $\hat{V}(x)$, we find (similar to the discrete case) the expected correspondence with wave-function notations:

$$\begin{aligned}\big\langle \Psi \,\big|\, \hat{V}(x) \,\big|\, \Psi\big\rangle &= \big\langle\Psi\big| \int_{-\infty}^{\infty} \mathrm{d}x'\big|x'\big\rangle\big\langle x'\big|\hat{V}(x) \int_{-\infty}^{\infty} \mathrm{d}x\, \big|x\big\rangle\big\langle x\big|\big|\Psi\big\rangle \\ &= \int_{-\infty}^{\infty} \mathrm{d}x' \int_{-\infty}^{\infty} \mathrm{d}x\, \langle \Psi \,|\, x'\rangle \underbrace{\big\langle x' \,\big|\, \hat{V}(x) \,\big|\, x\big\rangle}_{=V(x)\delta(x-x')} \langle x \,|\, \Psi\rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, \Psi^*(x)V(x)\Psi(x)\,,\end{aligned} \tag{5.7}$$

again, as expected.

Summarising:

Wavefunction $\Psi(x) \equiv \langle x \,|\, \Psi\rangle$ ***is the component of*** $\big|\Psi\big\rangle$ ***in the position eigenbasis*** $\big\{\big|x\big\rangle\big\}$***, while*** $\Psi^*(x) \equiv \langle \Psi \,|\, x\rangle$***;*** x ***is a 'continuous index' similar to*** n ***in discrete basis*** $\big\{\big|n\big\rangle\big\}$***;***

The bra(c)ket: $\langle \Psi \,|\, \Psi\rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, \Psi^*(x)\Psi(x)$, $\big\langle \Psi \,\big|\, \hat{V}(x) \,\big|\, \Psi\big\rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, \Psi(x)V(x)\Psi(x)$.

5.2 Position measurement

According to the 3rd QMP, measuring an observable $\hat{A}$ in a QM system in state $\big|\Psi\big\rangle$ yields one of its eigenvalues, A_n, with the probability $P_n = |\langle A_n \,|\, \Psi\rangle|^2$. The probabilities can be added: the probability that a measurement gives *either* A_n *or* A_m is given by the sum of the corresponding probabilities $P_{n\,\mathrm{or}\,m} = P_n + P_m$.

For the position operator, the probability to measure one of its eigenvalues is meaningless – the probability to get any particular value in continuum is zero. Hence we define the probability that position x is in certain range, $a < x < b$, *i.e.* take any value in the interval $[a, b]$. Generalising the sum of probabilities for discrete eigenvalues to continuum replaces, as always, the sum by the integral:

$$\mathrm{Prob}\,[a \leqslant x \leqslant b] = \int_a^b \mathrm{d}x |\langle x \,|\, \Psi\rangle|^2 = \int_a^b \mathrm{d}x |\Psi(x)|^2\,. \tag{5.8}$$

This is the probability to find the particle in the interval $[a, b]$ as was stated in QM2. The function $|\langle x \,|\, \Psi\rangle|^2 = |\Psi(x)|^2$ is the ***probability density*** for the position of the particle.

All the definitions above are generalised to a 3d space by changing $x \to \boldsymbol{r} = (x, y, z)$. The wave-function depends on three independent coordinates, $\Psi(x) \to \Psi(\boldsymbol{r}) = \langle \boldsymbol{r} \,|\, \Psi\rangle$, and the integration is carried out over volume element $\mathrm{d}\boldsymbol{r} \equiv \mathrm{d}^3x$ which in Cartesian coordinates is $\mathrm{d}\boldsymbol{r} \equiv \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$.

5.3 Momentum operator

Position operator is diagonal in its own eigenbasis, Eq. (5.4). What are matrix elements of momentum operator $\hat{p}$? Recall that in QM2 we used

$$\hat{p}\Psi(x) = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\Psi(x)\,. \tag{5.9}$$

With the new knowledge that $\Psi(x) = \langle x \,|\, \Psi\rangle$ we rewrite the left hand side as

$$\hat{p}\Psi(x) = \big\langle x\big|\hat{p}\big|\Psi\big\rangle\,, \tag{5.10}$$

i.e. the operator $\hat{p}$ acts on $\big|\Psi\big\rangle$, modifies it, and the x-component of the modified state $\big|\hat{p}\Psi\big\rangle$ is calculated. Let us insert the resolution of unity in the position basis:

$$\hat{p}\Psi(x) = \big\langle x\big|\hat{p}\big|\Psi\big\rangle = \int_{-\infty}^{\infty}\mathrm{d}x'\big\langle x\big|\hat{p}\big|x'\big\rangle\langle x' \,|\, \Psi\rangle = \int_{-\infty}^{\infty}\mathrm{d}x'\big\langle x\big|\hat{p}\big|x'\big\rangle\Psi(x')\,. \tag{5.11}$$

Comparing this with

$$-i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\Psi = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\int_{-\infty}^{\infty}\mathrm{d}x'\delta(x-x')\Psi(x') = \int_{-\infty}^{\infty}\mathrm{d}x'\left(-i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\delta(x-x')\right)\Psi(x'), \tag{5.12}$$

we identify the matrix element of the momentum operator in the position basis[1] $\big\{\big|x\big\rangle\big\}$:

$$\big\langle x\big|\hat{p}\big|x'\big\rangle = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\delta(x-x') \tag{5.13}$$

Unlike the position operator $\hat{x}$ diagonal in its eigenbasis, the momentum operator $\hat{p}$ is not: the difference between $\delta(x)$ and $\delta'(x)$ in the expressions for matrix elements is crucial!

Let us find the components of $\big|p\big\rangle$ in the position representation. We start with the EVE for $\hat{p}$ and *project* it into this representation by taking the inner product with $\big\langle x\big|$:

$$\hat{p}\big|p\big\rangle = p\big|p\big\rangle \quad \Rightarrow \quad \big\langle x\big|\hat{p}\big|p\big\rangle = p\langle x \,|\, p\rangle \equiv p\Psi_p(x)\,. \tag{5.14}$$

Here we identified the x-component of $\big|p\big\rangle$ with the wavefunction $\Psi_p(x)$, as Eq. (5.14) is a specific case of more general Eq. (5.10) with $\big|\Psi\big\rangle \to \big|p\big\rangle$. Using the resolution of unity and Eq. (5.13) we find

$$\big\langle x\big|\hat{p}\big|p\big\rangle = \int_{-\infty}^{\infty}\mathrm{d}x'\big\langle x\big|\hat{p}\big|x'\big\rangle\langle x' \,|\, p\rangle = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\int_{-\infty}^{\infty}\mathrm{d}x'\delta(x-x')\Psi_p(x) = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\Psi_p\,. \tag{5.15}$$

Hence we write the EVE (5.14) as a familiar differential equation whose solution is the plain wave:

$$-i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\Psi_p(x) = p\Psi_p(x) \quad \Rightarrow \quad \Psi_p(x) \equiv \langle x \,|\, p\rangle = A\mathrm{e}^{ipx/\hbar}\,. \tag{5.16}$$

The factor A should, in principle, be calculated from normalisation,

$$\langle p \,|\, p\rangle = \int \mathrm{d}x\langle p \,|\, x\rangle\langle x \,|\, p\rangle = \int \mathrm{d}x|\Psi_p(x)|^2 = |A|^2\int_{-\infty}^{\infty}\mathrm{d}x|\mathrm{e}^{ipx/\hbar}|^2. \tag{5.17}$$

[1] The derivative of Dirac delta function, $\delta'(x)$, is defined via its action on an arbitrary sufficiently smooth function $f(x)$ as $\int_{-\infty}^{\infty}\mathrm{d}x\,\delta'(x)f(x) = -\int_{-\infty}^{\infty}\mathrm{d}x\,\delta(x)f'(x) = -f'(0)$.

However, as $|\mathrm{e}^{i\alpha}| = 1$, this integral diverges: we already know from QM2 that the states belonging to continuous spectrum are not normalisable. Instead we specify coefficient A above by demanding the inner product $\langle p \,|\, p'\rangle = \delta(p - p')$, similar to that for position states,[2] Eq. (5.2)

$$\langle p' \,|\, p\rangle = |A|^2 \int_{-\infty}^{\infty} \mathrm{d}x \, \langle p' \,|\, x\rangle\langle x \,|\, p\rangle = |A|^2 \int_{-\infty}^{\infty} \mathrm{d}x \, \mathrm{e}^{i(p-p')x/\hbar} = 2\pi\hbar|A|^2\delta(p - p') \,. \tag{5.18}$$

Hence we choose $A = 1/\sqrt{2\pi\hbar}$ which determines the wavefunction of momentum eigenstate[3]:

$$\psi_p(x) = \langle x \,|\, p\rangle = \frac{1}{\sqrt{2\pi\hbar}} \mathrm{e}^{ipx/\hbar} \,. \tag{5.19}$$

The generalisation to 3D is straightforward: $p \to \boldsymbol{p}$, while $px \to \boldsymbol{p} \cdot \boldsymbol{r}$ so that the momentum operator, its matrix elements and its eigenfunctions in the position representation are

$$\hat{\boldsymbol{p}}\Psi(\boldsymbol{r}) = -i\hbar\nabla\Psi(\boldsymbol{r}) \,; \quad \left\langle \boldsymbol{r}\middle|\hat{\boldsymbol{p}}\middle|\boldsymbol{r}'\right\rangle = -i\hbar\nabla\delta(\boldsymbol{r} - \boldsymbol{r}') \,; \quad \Psi_{\boldsymbol{p}}(\boldsymbol{r}) = \langle \boldsymbol{r} \,|\, \boldsymbol{p}\rangle = \frac{1}{(2\pi\hbar)^{3/2}} \mathrm{e}^{i\boldsymbol{p}\cdot\boldsymbol{r}} \tag{5.20}$$

5.4 Momentum representation

The EVEs for x and p, Eqs. (5.4) and (5.14), are of the same structure. Hence, as we get any $\big|\Psi\big\rangle$ in the position representation by taking the inner product with $\big\langle x\big|$, we find $\big|\Psi\big\rangle$, and in particular $\big|x\big\rangle$, in the momentum representation by taking the inner product with $\big\langle p\big|$:

$$\langle p \,|\, \psi\rangle \equiv \Psi(p) \quad \Rightarrow \quad \psi_x(p) \equiv \langle p \,|\, x\rangle = \langle x \,|\, p\rangle^* = \psi_p^*(x) = \frac{1}{\sqrt{2\pi\hbar}} \mathrm{e}^{-ipx/\hbar} \,. \tag{5.21}$$

Note that $\Psi(p) = \langle p \,|\, \Psi\rangle$ plays the same role in the momentum representation as the wave function $\Psi(x) = \langle x \,|\, \Psi\rangle$ in the position representation. In particular, $|\Psi(p)|^2$ is the probability density for momentum, *i.e.* can be used to calculate the probability

$$\mathrm{Prob}\,[p_a \leqslant p \leqslant p_b] = \int_{p_a}^{p_b} \mathrm{d}p \, |\Psi(p)|^2 \tag{5.22}$$

to find a quantum particle in state $\big|\Psi\big\rangle$ in the momentum interval between p_a and p_b.

The wavefunctions in momentum and position representations are related by the Fourier transform, as we find by using the resolution of unity in both representations[4]:

$$\Psi(p) = \langle p \,|\, \Psi\rangle = \int_{-\infty}^{\infty} \mathrm{d}x \, \langle p \,|\, x\rangle\langle x \,|\, \Psi\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \mathrm{d}x \, \mathrm{e}^{-ipx/\hbar}\Psi(x) \tag{5.23}$$

$$\Psi(x) = \langle x \,|\, \Psi\rangle = \int_{-\infty}^{\infty} \mathrm{d}p \, \langle x \,|\, p\rangle\langle p \,|\, \Psi\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \mathrm{d}p \, \mathrm{e}^{ipx/\hbar}\Psi(p) \,. \tag{5.24}$$

[2] Reminder from Maths2: $\int_{-\infty}^{\infty} \mathrm{d}x \, \mathrm{e}^{ikx} = 2\pi\delta(k)$; then, $\delta(ak) = (1/|a|)\delta(k)$.

[3] Such a choice of A is not always convenient, since the WF $\psi_p(x)$ is still *not normalised*. Hence, it cannot be directly substituted when counted probabilities, since in the 3rd QMP all states assumed to be normalised. For example, the probability of a free particle to be in any particular interval $[a, b]$ in the *infinite* system is always 0 but if one substituted $\Psi_p(x)$ of Eq. (5.19) into Eq. (5.8), the answer would be finite, and of a wrong dimensionality. The way out is to consider a system of finite large length, when the momentum becomes discrete – a QM particle in a rectangular box with the normalisable WF.

[4] Here the choice of A in Eq. (5.18) and hence in Eq. (5.21) imposed equal coefficients in the direct and inverse FT.

Find dimensions of $\Psi(x)$ and $\Psi(p)$ and explain why they differ from those of $\psi_p(x)$ and $\psi_x(p)$.

In physical terms one notices that a state with well-defined position x is a linear superposition of infinitely many states with well defined momentum and vice versa. This is a consequence of the celebrated Heisenberg uncertainty principle, to which we shall return soon.

Note also that the completeness of the momentum eigenbasis (the resolution of unity),

$$\int_{-\infty}^{\infty} \mathrm{d}p|p\rangle\langle p| = \hat{I} \tag{5.25}$$

used in Eq. (5.24), follows from that in the position eigenbasis, Eq. (5.2) which can be demonstrated by taking the position matrix elements of the l.h.s. and r.h.s. of Eq. (5.25):

$$\text{l.h.s.: } \langle x'| \left(\int_{-\infty}^{\infty} \mathrm{d}p|p\rangle\langle p| \right) |x\rangle = \int_{-\infty}^{\infty} \mathrm{d}p \langle x' \,|\, p\rangle \langle p \,|\, x\rangle = \int_{-\infty}^{\infty} \frac{\mathrm{d}p}{2\pi\hbar} \mathrm{e}^{ip(x-x')/\hbar} = \delta(x'-x)$$

$$\text{r.h.s.: } \langle x'|\hat{I}|x\rangle = \langle x' \,|\, x\rangle = \delta(x'-x)\,.$$

Lecture 6. Heisenberg uncertainty principle

Here, after introducing spatial translations, we re-iterate that two observables can be measured simultaneously only if the appropriate operators commute, and go from that to the HUP.

6.1 Momentum operator and spatial translations

Momentum is related to translations. To see this we consider the unitary operator[1]

$$\hat{T}(a) = \mathrm{e}^{-ia\hat{p}/\hbar} = 1 - \frac{i}{\hbar}a\hat{p} - \frac{1}{2\hbar^2}a^2\hat{p}^2 + \dots\,, \tag{6.1}$$

where a is some length. Acting by $\hat{T}$ on position eigenket $\big|x\big\rangle$, we find:

$$\hat{T}(a)\big|x\big\rangle = \hat{I}\hat{T}(a)\big|x\big\rangle = \int_{-\infty}^{\infty}\mathrm{d}p\big|p\big\rangle\big\langle p\big|\hat{T}(a)\big|x\big\rangle = \int_{-\infty}^{\infty}\mathrm{d}p\big|p\big\rangle\big\langle p\big|\mathrm{e}^{-ia\hat{p}/\hbar}\big|x\big\rangle\,, \tag{6.2}$$

where the p-resolution of unity was used again.

Since $\hat{p}$ is Hermitian, $\hat{p} = \hat{p}^\dagger$, it can act on the bra as well as on the ket, giving the EVE $\big\langle p\big|\hat{p} = p\big\langle p\big|$. As we know from maths, $\big\langle p\big|f(\hat{p}) = f(p)\big\langle p\big|$ for any smooth function f. Hence we get

$$\begin{aligned}&\big\langle p\big|\mathrm{e}^{-ia\hat{p}/\hbar}\big|x\big\rangle = \mathrm{e}^{-ipa/\hbar}\langle p\,|\,x\rangle = \frac{1}{\sqrt{2\pi\hbar}}\mathrm{e}^{-ip(x+a)/\hbar} = \langle p\,|\,x+a\rangle\\ \Rightarrow\quad &\hat{T}(a)\big|x\big\rangle = \int_{-\infty}^{\infty}\mathrm{d}p\big|p\big\rangle\langle p\,|\,x+a\rangle = \big|x+a\big\rangle\end{aligned} \tag{6.3}$$

Therefore, the action of $\hat{T}(a)$ on $\big|x\big\rangle$ is the translation $\big|x\big\rangle \to \big|x+a\big\rangle$.

6.2 Noncommuting variables & generalised Heisenberg uncertainty principle

The state $\big|x+a\big\rangle$ is an eigenstate of $\hat{x}$ corresponding to the eigenvalue $x+a$. Hence

$$\hat{x}\hat{T}(a)\big|x\big\rangle = \hat{x}\big|x+a\big\rangle = (x+a)\big|x+a\big\rangle\,. \tag{6.4}$$

Had we applied the position operator *before* acting with $\hat{T}(a)$ the result would be different:

$$\hat{T}(a)\hat{x}\big|x\big\rangle = \hat{T}(a)x\big|x\big\rangle = x\hat{T}(a)\big|x\big\rangle = x\big|x+a\big\rangle\,. \tag{6.5}$$

Subtracting Eq. (6.5) from Eq. (6.4), we find the commutator

$$\big[\hat{x},\hat{T}(a)\big]\big|x\big\rangle = a\big|x+a\big\rangle = a\hat{T}(a)\big|x\big\rangle \quad\Rightarrow\quad \big[\hat{x},\hat{T}(a)\big] = a\hat{T}(a)\,. \tag{6.6}$$

When displacement $a \to 0$, it follows from expansion (6.1) that the r.h.s. of Eq. (6.6), $a\hat{T}(a) \simeq a$, while the l.h.s. is given by

$$\big[\hat{x},\hat{T}(a)\big] \simeq \big[\hat{x}, 1 - i\frac{a\hat{p}}{\hbar}\big] = -\frac{ia}{\hbar}\big[\hat{x},\hat{p}\big] \text{ so that } \frac{a}{i\hbar}\big[\hat{x},\hat{p}\big] = a \quad\Rightarrow\quad [\hat{x},\hat{p}] = i\hbar\,. \tag{6.7}$$

[1] Recall that $\mathrm{e}^{i\hat{h}}$ is a unitary operator for any Hermitian $\hat{h}$, Eq. (2.32).

Hence we have recovered the familiar commutation relations between position and momentum, from which (as you know from QM2) follows the Heisenberg uncertainty principle (HUP),

$$\Delta x \Delta p \geqslant \hbar/2. \tag{6.8}$$

We shall now extend the HUP to any pair of non-commuting Hermitian operators. First, we define formally the uncertainty, ΔA, in measuring an observable $\hat{A}$ in a normalised state $\big|\Psi\big\rangle$ as the square root of the *standard deviation* or *variance*:

$$(\Delta A)^2 = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2 = \big\langle \Psi \big| \hat{A}^2 \big| \Psi \big\rangle - \big\langle \Psi \big| \hat{A} \big| \Psi \big\rangle^2 . \tag{6.9}$$

The variance $\langle \hat{A}^2 \rangle$ and the expectation value $\langle \hat{A} \rangle$ characterise the statistical distribution of results of repeated measurements of $\hat{A}$: the smaller ΔA, the more precisely one can know $\hat{A}$.

Let me stress here that the statistical averaging over many measurements, like $\langle \hat{A} \rangle$, and the QM expectation value in a given state, $\big\langle \Psi \,\big|\, \hat{A} \,\big|\, \Psi \big\rangle$, are directly connected. Indeed, using the resolution of unity in the eigenbasis $\big\{ \big| n \big\rangle \big\}$ of $\hat{A}$ defined by the EVE, $\hat{A}\big| n \big\rangle = a_n \big| n \big\rangle$, we find

$$\big\langle \Psi \big| \hat{A} \big| \Psi \big\rangle = \sum_n a_n |\langle n \,|\, \Psi \rangle|^2 = \sum_n a_n P_n, \text{ and } \big\langle \Psi \big| \hat{A}^2 \big| \Psi \big\rangle = \sum_n a_n^2 |\langle n \,|\, \Psi \rangle|^2 = \sum_n a_n^2 P_n \tag{6.10}$$

so that

$$(\Delta A)^2 = \sum_n a_n^2 P_n - \sum_{m,n} a_m P_m a_n P_n = \langle a_n^2 \rangle - \langle a_n \rangle^2 = \langle a_n - \langle a_n \rangle \rangle^2 \tag{6.11}$$

is indeed the standard deviation of a random variable taking values a_n with probabilities P_n.

The HUP, Eq. (6.8), is generalised to any pair of Hermitian operators $\hat{A}$ and $\hat{B}$ as[2]

$$\Delta A \Delta B \geqslant \tfrac{1}{2} \Big| \big\langle \big[\hat{A}, \hat{B} \big] \big\rangle \Big| = \tfrac{1}{2} \Big| \big\langle \Psi \big| [\hat{A}, \hat{B}] \big| \Psi \big\rangle \Big| \to \tfrac{1}{2} |c|, \tag{6.12}$$

the latter being valid if the commutator $\big[\hat{A}, \hat{B} \big]$ is a c-number. As this is true for $\big[\hat{x}, \hat{p} \big]$, in this case Eq. (6.12) reduces to Eq. (6.8). For two general noncommuting operators, the r.h.s. remains proportional to the expectation value of their commutator.

If the commutator of two observables vanishes, $[\hat{A}, \hat{B}] = 0$, they are said to be *compatible*. As it was shown in Maths2B, such operators can be diagonalised simultaneously, *i.e.* they share the same eigenbasis, say $\big\{ \big| n \big\rangle \big\}$, with different eigenvalues:

$$\hat{A}\big| n \big\rangle = a_n \big| n \big\rangle; \qquad\qquad \hat{B}\big| n \big\rangle = b_n \big| n \big\rangle. \tag{6.13}$$

In this case, if a measurement of $\hat{A}$ resulted in a_m, then the system collapses to state $\big| m \big\rangle$ so that an immediate measurement of $\hat{B}$ will give result b_m *with probability 1*. This is the meaning of the statement that compatible observables can be measured simultaneously.

[2] One of many possible variants of proving this is given at the end of this lecture.

On the contrary, if commutator $[\hat{A},\hat{B}] \neq 0$, the observables are *incompatible* and the common eigenbasis cannot be found. In this case, when measuring $\hat{A}$ returns a_n, $\big|\Psi\big\rangle$ collapses to the eigenstate $\big|a_n\big\rangle$. This is not an eigenstate of $\hat{B}$, defined by the appropriate EVE, $\hat{B}\big|b_n\big\rangle = b_n\big|b_n\big\rangle$, but rather a linear superposition of such eigenstates:

$$\big|a_n\big\rangle = \sum_n \alpha_n \big|b_n\big\rangle\,, \text{ with } \alpha_n = \langle b_n \,|\, a_n\rangle. \tag{6.14}$$

Hence, according to the 3$^{\rm rd}$ QMP, a measurement of $\hat{B}$ would give any of its eigenvalues b_n with the probability $P_{b_n} = |\alpha_n|^2$.

It might happen that $\big[\hat{A},\hat{B}\big]$, being nonzero in general, has zero expectation value for a particular choice of the state, *i.e.* for a certain $\big|\Psi_0\big\rangle$ we have $\big\langle\Psi_0\big|\big[\hat{A},\hat{B}\big]\big|\Psi_0\big\rangle = 0$. In this case the HUP allows a simultaneous measurement of incompatible observables. Let us look, as an example, at three components of angular momentum, $\hat{\boldsymbol{L}} = \hat{\boldsymbol{r}} \times \hat{\boldsymbol{p}} = (\hat{L}_x, \hat{L}_y, \hat{L}_z)$. The commutator

$$[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z\,, \tag{6.15}$$

vanishes for the state corresponding to zero eigenvalue of $\hat{L}_z$. Then there is a *possibility* to measure all the three components of angular momentum simultaneously but only in the case when the total angular momentum is also zero. Then all the three components of $\hat{\boldsymbol{L}}$ vanish.

6.2.1 A proof of the generalised HUP

Now we prove the generalised HUP of Eq. (6.12). First we shift operators $\hat{A}$ and $\hat{B}$ so that they have zero expectation values in the state $\big|\Psi\big\rangle$:

$$\hat{A} \to \hat{A} - \big\langle\Psi\big|\hat{A}\big|\Psi\big\rangle \qquad \hat{B} \to \hat{B} - \big\langle\Psi\big|\hat{B}\big|\Psi\big\rangle \tag{6.16}$$

Thus the uncertainties reduce to $(\Delta A)^2 = \big\langle\Psi\big|\hat{A}^2\big|\Psi\big\rangle$ and $(\Delta B)^2 = \big\langle\Psi\big|\hat{B}^2\big|\Psi\big\rangle$. Consider the state

$$\big|\Phi\big\rangle = \hat{A}\big|\Psi\big\rangle + i\lambda\hat{B}\big|\Psi\big\rangle = \big|\big(\hat{A}+i\lambda\hat{B}\big)\Psi\big\rangle\,, \tag{6.17}$$

where λ is a real number. Writing the inner product, $\langle\Phi \,|\, \Phi\rangle \geqslant 0$, using Eq. (6.17) and the fact that both operators are Hermitian, *i.e.* $\big\langle\Phi\big| = \big\langle\Psi\big(\hat{A}-i\lambda\hat{B}\big)\big|$, we find

$$\langle\Psi\big(\hat{A}-i\lambda\hat{B}\big) \,|\, \big(\hat{A}+i\lambda\hat{B}\big)\Psi\rangle = \big\langle\Psi\big|\hat{A}^2\big|\Psi\big\rangle + i\lambda\big\langle\Psi\big|[\hat{A},\hat{B}]\big|\Psi\big\rangle + \lambda^2\big\langle\Psi\big|\hat{B}^2\big|\Psi\big\rangle \geqslant 0 \tag{6.18}$$

The commutator of Hermitian operators is anti-Hermitian (hence its expectation value is purely imaginary) as $[\hat{A},\hat{B}]^\dagger = (\hat{A}\hat{B} - \hat{B}\hat{A})^\dagger = \hat{B}^\dagger\hat{A}^\dagger - \hat{A}^\dagger\hat{B}^\dagger = [\hat{B},\hat{A}] = -[\hat{A},\hat{B}]$. Therefore, one writes

$$(\Delta A)^2 - \lambda\Big|\big\langle\big[\hat{A},\hat{B}\big]\big\rangle\Big| + \lambda^2(\Delta B)^2 \equiv a\lambda^2 + b\lambda + c \geqslant 0\,. \tag{6.19}$$

The condition for the quadratic function of λ to be non-negative for all λ is to have no real roots, *i.e.* to have the negative discriminant

$$D = b^2 - 4ac = \Big|\big\langle\big[\hat{A},\hat{B}\big]\big\rangle\Big|^2 - 4\Delta A\Delta B \leqslant 0\,. \tag{6.20}$$

which is nothing else but the generalised HUP of Eq. (6.12).